

**Research &
Statistics in
NURSING**
& Related Health Fields:
A Practical Approach



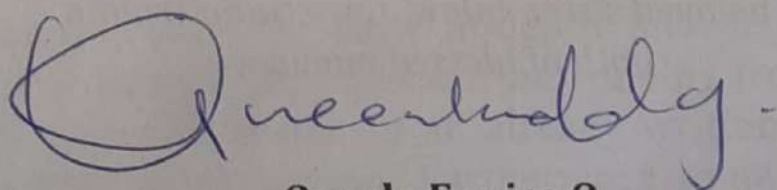
Osuala E.O

Second Edition, 2021

***Research & Statistics
In Nursing & Health Related Fields:
A Practical Approach
(Includes index)***

OSUALA EUNICE O. , RN, RM, MPH, PhD ,FWACN
*Head of Department of Nursing Science
PAMO University of Medical Sciences,
Port-Harcourt*

*Former Head of Department of Nursing Science
Department of Nursing Sciences
Nnamdi Azikiwe University*



Osuala Eunice O.

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CHAPTER 1

INTRODUCTION

It has been appreciated that nurses, students of various aspects of health, like other professionals will be involved in research, either their own or someone else's hence the introduction of research study at all levels of endeavor. Research is time consuming, as one needs time to develop skill through practice. For a preliminary course in research methodology to be meaningful, one is expected to carry out a small study through which the research process could be learned and understood. Aina says that, from all indications it seems nurses have a fear of research. In order to overcome the research phobia, this handbook for students studying nursing and related fields was provided.

DEFINITION OF RESEARCH

The word research, though in common usage often-mean different things to different individuals. Akpala sees it as a systematic search for new facts, knowledge, getting additional information and search for the truth. Barnes says that research is a way of dealing with ideas, while Notter in Abdellah says research systematically searches for new facts and relationship. Man is an inquisitive animal with quest for new knowledge and absolute truth, wanting to answer the questions; how, when and why about things around him. The author therefore defines Research as the identification of problem and making predictions or explanations through a planned and systematic collection of data geared towards the provision of answer(s) to problem (s). Nursing research covers all aspects of health and illness as well as associated events in the society among individuals and populations.

PURPOSE OF RESEARCH

Generally, research has many purposes. The knowledge of research has numerous advantages since it will go a long way in helping to find the absolute truth. Several reasons are provided here why individuals and establishments are carrying out research.

One will develop the research skill by carrying out a study. Anyone involved in research, may have to read many write ups and by so doing, will improve one's ability to quickly scan through volumes of documents and pick the relevant and useful information needed for the study. It helps one discover various exciting facts; why do tuberculosis bacilli cause pulmonary tuberculosis? Why does the sun rise in the east and set in the west? Why is the weather harsh in recent times especially in the tropics? How did the Universe come into existence and true meaning of human life and why do people die? Why should certain people be affected by a particular ailment when others around them will not be affected? Research answers all these questions. Explaining death as part of living is a purpose of research. Research will help in upgrading existing methods of dealing with certain problems or issues. For instance, the ingenuity in changing the manual gear system of a car to automatic gear makes driving a lot easier and encourages many people to drive, even the aged and handicapped. Record keeping has been made easy through computerization. Records of events that happened many years ago can be retrieved within seconds using the internet. Airbed and waterbed takes off the problem of bed sore. Vaccines for prevention of measles, tuberculosis (BCG), and cervical cancer (Cervarix) etc. are outcomes of researches. The guidelines for prevention, promotion, maintenance of health and even cure for many communicable and non-communicable diseases emanate as a result of scientific research studies. Intensive studies are ongoing in the world today in search of preventive vaccine for COVID- 19 and its possible cure. Extracts from plants serve as drugs and Pharmacists

under the specialty of Pharmacognosy are well grounded in this area. All these inventions came about through research.

Through research, pathways of action of known substances or new elements are discovered in order to improve human life and improve longevity. Substances like alcohol and formalin help in preservation. Corpses can therefore be preserved and disposed at the survivor's convenience. Fluorine, which is added to water makes it safe for drinking and limits water borne diseases. Based on research, events can be predicted. Weather forecast by meteorologist is based on research. Situations and events can be brought under control as a result of research. An outbreak of cholera in a community is brought under control through epidemiological approach. Associations are proved through research. People that smoke for a prolonged period are liable to lung cancer, thus the Federal Ministry of Health, warns that tobacco smoking is dangerous. This has been proved through both retrospective and prospective studies, thereby precipitating the warning.

Research is needed to improve nursing care for the good of all as well as for professional development. Nurses will have to make decisions in many areas, for example, how to deal with colleagues. Research is needed to know how to relate well with the public, to conserve money, time and energy during practice. We must find out why, when and how things go wrong or well with the profession and this is done through research. The concept of Evidence Based Practice is a product of research. Keying into world's best practices by nurses cannot be achieved and sustained without scientific backing from research findings. More theories especially in nursing need to be developed. Theories applied to nursing are almost all derived from other fields as medicine, pharmacology and social sciences. Research is also needed to develop a body of knowledge specific to nursing. In order to identify nursing needs in the areas of education and practice, research must be carried out. The role of a

nurse seems to be undefined more so with technology explosion and erosion of the professional boundaries. Nursing has been recognized as a profession, both in developed and developing countries; it needs research to define its area or scope of operation in order to have full autonomy in area of practice.

Research enables health professionals to acquire new knowledge on diseases process and its causes, in addition to new intervention and technique required to promote health and prevent or cure diseases, to improve and maintain the health status of both individuals and community as a whole. It contributes in various aspects of the profession as midwifery. The anti-shock garment which helps in the management of post-partum hemorrhage as well as Family Planning which helps in population control and improved resources and longevity was discovered through research. In decision making, research makes for better judgment in our day-to-day activities and helps to improve our thinking process through critical assessment of issues. Research not only exposes the problem at stake, but also the gravity. It helps in National planning and building. It makes for both self and professional growth. Research is a "stitch in time that saves nine". It is an imperative in the achievement of the Sustainable Development Goals.

AREAS OPEN TO RESEARCH IN NURSING

Research topics can be derived from teachers, organizations, literature or from students themselves. Whatever research topic one adopts, it is usually influenced by one's area of specialization, concern or interest. Many areas are open to research in Health such as the incidence or prevalence of disease or event (hypertension, post-partum hemorrhage and accidents etc). How resources could be distributed to meet the demand and time can be researched on. 'Adequacy' of nursing care is yet to be defined. What should the role of a nurse be? The ecology of nursing, how many nurses do we have? Is the number deficient, enough or in excess? What are their ranks

and specialties? Are biological, behavioral, social sciences relevant to nursing? All these are researchable topics. The Awareness, knowledge, perception, attitude and practice of group of people on typical health and health related issues need to be researched. Specific examples include; child abuse, domestic violence, road traffic accident, Breast self-examination, infection control. Parental awareness on causes, effect of diseases on the child's health can be investigated on. The causes, effects and preventive measures of HIV/AIDS amongst our youths, pregnant women, etc need to be investigated. Factors responsible for non-use of nursing process in the care of patients in various hospitals across the nation, factors responsible and incidence of various infectious diseases among children or certain vulnerable groups in the community (gastroenteritis, pulmonary tuberculosis, etc) are open to research.

Other areas include effects of traditional belief and culture on the health of children, pregnant women, females, widows etc in the community. Utilization of health services by individuals, groups and communities, such as use of immunization services by mothers can be verified. Effects of alcohol consumption on men of certain age and the youth; parental awareness on causes, effect and prevention of anemia; factors influencing the choice of nursing as a profession by males; poverty, its causes and consequences, all need to be investigated. More topics can be developed based on attitude of:

- Doctors and Nurses towards patients.
- People of various ages and groups towards test for HIV and Covid-19.
- People towards care of psychiatric relatives.
- Male adults towards vasectomy.
- Mothers towards use of insecticide nets, Oral Rehydration Therapy, Exclusive Breast Feeding, Family Planning methods.
- Nurses towards Geriatrics and patients with terminal illness etc.

MEANING OF RESEARCH

Even though research has already been defined, using the word as an acronym, it stands thus:

- Review Literature
- Evolve a theoretical framework
- State the problem or purpose and hypothesis
- Evolve a methodology for data collection
- Analyze and interpret the data
- Report conclusion and recommendation
- Communicate result
- Help to implement finding

Note these are steps involved in research i.e. research process.

1. THEORIES, MODELS AND CONCEPTUAL FRAMEWORKS

At the stage of literature review, students are required to provide information under subheadings such as theoretical, Empirical reviews and conceptual framework. Theoretical review takes into account the theories or information on the variable(s) being investigated as found in text books, monographs and the like. A student investigating a particular disease may give information on its causes, signs and symptoms, prevention, etc. as gathered from various texts. Empirical review focuses on the studies done by various authors /researchers on the variable(s) under study. It may include works done, regionally, nationally or even within one's State or Local Government Area. If there is paucity of data on the topic being investigated, the search may go global. Conceptual framework involve translating the theories or models found suitable for the study into a sketch that shows a link between the dependent, intervening and independent variables for an output or result to evolve.

Purpose in research are to influence the following:

- i. Phenomenon to be studied
- ii. The research techniques and tools to be used.
- iii. The precise nature of the problem to be studied or the purpose to be fulfilled by the investigator or both
- iv. Settings in which data is to be gathered and the subjects who are to provide the data.
- v. The methods of data collection and analysis
- vi. Contribution that the research would make to the advancement of knowledge. (Verhonick, PJ {Ed} 1975.p.7)

Theory is a proven fact. It involves a creative and rigorous structuring of ideas that project a tentative, purposeful, and systematic view of phenomena. Through systematic inquiry, nurses are able to develop knowledge relevant to improving the care of patients.

A model is a way of presenting a situation in such a way that it shows the logical terms in order to showcase the structure of the original idea.

A framework usually denotes a structure, overview, outline, system or plan consisting of various descriptive categories, e.g. concepts, constructs or variables, and the relations between them that are presumed to account for a phenomenon.

Conceptual framework deals with abstractions which are assembled together by virtue of their relevance to the research problem. It provides the researcher with the evidence needed to justify why the particular problem is chosen or selected. Both

theoretical and conceptual framework are at times used synonymously in research.

Theoretical Framework

When a study is based on a theoretical framework, it adds weight to findings. A theory is an abstract generalization that explains systematically the relationships among phenomena. The overall objective of theory is to make scientific findings meaningful and generalizable. Theories also help to summarize existing knowledge into coherent sets of facts. Examples of the theories on which Nursing research are based on, are subsequently provided.

a. Social Learning theory:

A psychologist, Julian Rotter postulated that human behavior in specific situations are contingent upon the individual's expectancy that the particular behavior will be rewarded. A research may focus on the relationship between action or input and output or reward. This theory has generated extensive research interest in many disciplines, nursing inclusive. Most theories and models adapted in nursing research are from the field of psychology and this is why works by people like Banduras, Jean Piaget, Pavlov and Maslow are commonly applied to nursing education, practice and research. Albert Bandura was one of the greatest psychologist and researcher of our time. He is best known for developing the social learning theory. In his experiments, Bandura sought to better understand how children learn and express emotions and behavior. Other researchers continue to consider his theories and experiments as they learn new information about social learning and behavior. Social Learning theorists explain behavior in terms of interaction between cognitive, behavioral & environmental determinants.

Bandura consider learning principles to be sufficient to explain and predict behavior and behavior change. The focus is on interaction, between the external stimuli and internal cognition, in a social

context. He suggested that most human behavior is learned by observation. In Modeling we observe the behavior of others and use the information as a guide for our own behavior. Models provide actions and behavior for children to emulate. Children are surrounded by people for them to emulate—their parents, friends, teachers, characters on television and others. The people and characters in their lives provide models of behavior for them to imitate.

The social learning theory suggests that children are more inclined to imitate those whom they believe are similar to themselves, as in the same gender. The concept is that people around children will respond to children's behavior by reinforcing it or by punishing it. Children will repeat behavior that others reinforce and adults can strengthen good behavior by reinforcing it. Reinforcements can be internal or external and positive or negative.

Social learning theory postulates that human behaviours in specific situations are contingent upon the individual's expectancy that a particular behavior will be reinforced (rewarded). This theory can be used as a basis for an investigation of achievement behavior in some group of a people. Eg. It can be used to judge the degree to which chronically ill children believe that their efforts bring about success or failure. A comparison can be made between the ill and healthy children.

b. Kinesthetic Need Theory:

This was developed by Kulka, who postulates that infants have kinesthetic needs or drives (incoming sensory stimulations) which compel the infant to move. This is in an effort to relieve the tension and return the body to a state of equilibrium. One can conduct a study based on this theory by finding out effect of increased stimulation on the child. This can be done using experimental research design, allocating subjects into control and experimental groups.

Cognitive Dissonance Theory:

Festinger postulates that the presence of dissonance motivates a person both to seek consonance and to avoid situations that will likely increase the dissonance. Consonance implies mutual agreement between some aspects of a person's experiences. A researcher can use this theory as a basis for investigating health beliefs in some disease conditions. If the disease condition is made the dependent variable, and the perceived benefits from some actions of the individual is made the independent variables, a relationship can be established between those who are susceptible and those who are not. Eg. Stillman in 1977 used this theory when a study was conducted on women who perceived themselves as more susceptible to breast cancer and those less susceptible. The women that were more susceptible carried out breast examination for cancer more often than the women who perceived themselves as less susceptible.

d. Nursing practice theories

Nursing theories though not many, can be applied to many different circumstances, even to research. Nursing practice theories have the most limited scope and level of abstraction and are developed for use within a specific range of nursing situations. Nursing practice theories provide frameworks for nursing interventions, and predict outcomes and the impact of nursing practice. Nursing practice theories are usually defined to an exact community or discipline. These are major Nurse theorists whose works are applied most often in nursing practice and research.

Major nursing theorists and their theories are:

- Martha E. Rogers : Science of unitary human beings
- Madeleine Leininger : Culture Care Theory
- Hildegard Peplau : Theory of interpersonal relations
- Josephine Paterson and Loretta Zderad: Humanistic Nursing Theory

- Jean Watson: Human caring theory
- Nancy Roper : Activities of Daily Living theory
- Helen Erickson: Modeling & Role modeling theory
- Virginia Henderson: Henderson's need theory
- Anne Casey: Casey's model of nursing
- Betty Neuman: Neuman systems model
- Callista Roy: Adaptation model of nursing
- Merle Mishel: Uncertainty of Illness Theory
- Faye Abdellah: Patient-centered approach to Nursing
- Dorothea Orem: Self-care deficit nursing theory

Dorothea Orem-Self-Care Nursing Theory

Dorothea Orem, the Self-Care Nursing Theory (also known as the Self-Care Deficit Nursing Theory) focuses on the nurse's role in supporting the patients' ability to be self-sufficient and responsible for their own care. The theory is based on the idea that people must be knowledgeable about their health problems to provide adequate self-care. The theory is made up of three interconnected theories: the theory of self-care, the theory of self-care deficit and the theory of nursing systems. Orem based her theory on her practice as a nurse and contemporary nursing literature and thought.

Leininger -Culture Care Theory

Madeleine Leininger recognized a lack of cultural and care knowledge as a missing component in nursing and patient care. She developed transcultural nursing and the Cultural Care Theory to study and explain the outcomes of transcultural nursing. Leininger's theory is focused on providing care that is in harmony with a patient's cultural beliefs, practices and values. In the 1960s she coined the phrase "culturally congruent care," which is the primary goal of transcultural nursing.

Selected theories and models

Theoretical reviews

Example 1: The theoretical model selected for this example was Bandura's (1986) social cognitive theory. According to Bandura (1997) social cognitive theory presents a model "of interaction in which the events of the environment and personal behavior act as reciprocal determinant factors of one another". The theory places emphasis on auto direction, autoregulation, and perception of self-efficacy. According to Glanz, Rimer, and Lewis (2002), social cognitive theory proposes that a human being possesses the following capabilities: (i) symbolization (processing and transformation of short term experiences into internal models that serve as guidelines for future actions), (ii) advanced thought (the ability to anticipate consequences, which is similar to future actions and future planning of courses of action to attain valued goals), (iii) vicarious learning (acquiring rules for generation of action to attain valued goals), (iv) auto-regulation (using internal rules and the auto-evaluation of reactions as a means of encouraging and regulating the behavior and fixing the external environment to create incentives for action), and (v) auto-reflection (thinking of one's own process of thinking and actively modifying the thoughts). Given these capabilities, human behavior is shaped by the interaction between acquiring knowledge, gaining experience, and the individual's environment. The social cognitive theory is characterized by reciprocal determinism. Reciprocal determinism refers to the notion that cognition (perceived ability to perform the task), environment (the setting), and behavior (the task being performed) are bidirectional; i.e., the components influence and are influenced by one another. The degree of influence that each of the three components has is contingent upon the type of activity, the individual and the situation (Bandura, 1989).

According to social cognitive theory, beliefs are formed through observation and reflective thinking. These beliefs include self-attribution, self-assessment, and self-efficacy. A person's perceived self-efficacy is a belief in their abilities to carry out a particular

action. Opinions concerning self-efficacy can develop through experiences, learning, verbal persuasion, and somatic responses to particular situations. Low self-esteem can result in failure and a lack of competition and growth. Belief in themselves makes it possible for individuals to complete tasks that build their competence and confidence. A greater perceived efficacy can cause more vigorous and persistent individuals to engage actively in a particular behavior, even if they face obstacles and adverse experiences (Bandura, 1997).

The ability of learners to acquire a knowledge base, self-administration, and interpersonal skills required to succeed in an occupation is partially determined by their perception of their own efficacy.

The promotion of knowledge, experience, achievements, and support systems comes before the acquisition of self-confidence. Eventually, graduates from nursing programs will enter the professional nursing field with high levels of self-confidence. This will allow nurses to develop: (i) clinical performance, (ii) the ability to produce change, (iii) the power to execute tasks or handle situations that test their own skills, (iv) professional collaboration, and (v) the ability to motivate others. The resulting self-confidence among nurses will be beneficial to the consumer of the nursing care as well as his or her families (White, 2009).

Application of the theory to the study.

Behavior is the outcome of the interaction between cognition, modeling, practice, and feedback, which develops self-efficacy and the ability to identify and enlist social supports in the environment (Bandura, 1986). Along with cognitive information, modeling is an effective method for developing the social and self-regulative skills needed to translate information into action (Bandura, 1994). As Bandura (1997) stated, "after the capacity for observational learning has fully developed, one cannot keep people from learning what they have seen". By taking students in an environment where Breast self-examination (BSE) technique is the norm and demonstrating

how to promote and teach the BSE skill to patients, knowledge will be more easily transformed into behavior.

According to Bandura (1986), another component of an effective behavioral program is skill enhancement to develop resilient self-efficacy, which is the perceived ability and confidence to perform a specific behavior (Bandura, 1986). Providing opportunities for guided practice and consistent feedback helps to initiate the development of self-efficacy and confidence. Nursing faculty need to plan clinical time for student practice and return demonstration with feedback. BSE is a learned skill for the student, patient, and the health professional care giver. The final component is that of identifying and enlisting social supports for the individual desired change. These will give the individual the requisite for maintaining his or her behavior.

Example II: The conceptual framework used for this study was derived from the theory of planned behavior and role theory.

The theory of planned behavior

The theory of planned behaviour is about the link between attitudes and behavior. It was proposed by Icek Ajzen in 1985 as an extension of the theory of reasoned action. It is one of the most predictive persuasion theories. It has been applied to studies of the relations among beliefs, attitudes, behavioural intentions and behaviors in various fields such as advertising, public relations, campaigns, healthcare, etc. The theory was developed from the Theory of Reasoned Action, which was proposed by Martin Fishbein together with Icek Ajzen in 1975. According to the Theory of Reasoned Action, if people evaluated the suggested behavior as positive (attitude), and if they think their significant others wanted them to perform the behavior (subjective norm), this results in a higher intention (motivation) and they are more likely to do so. A high correlation of attitudes and subjective norms to behavioral intention, and subsequently to behavior has been confirmed in many studies (Ajzen, 1985).

Since behavioral intention cannot be the exclusive determinant of behavior where an individual's control over the behavior is incomplete, Ajzen introduced the Theory of Planned Behavior by adding a new component, "perceived behavioral control." By this, he extended the Theory of Reasoned Action to cover volitional behaviors for predicting behavioral intention and actual behavior. In addition to attitudes and subjective norms the additional concept of the Theory of Planned Behavior is perceived behavioral control, which originates from Self Efficacy Theory (SET).

Self-efficacy is the conviction that one can successfully execute the behavior required to produce the outcomes. The outcome expectancy refers to a person's estimation that a given behavior will lead to certain outcomes. He states that self-efficacy is the most important precondition for behavioural change, since it determines the actual behavior.

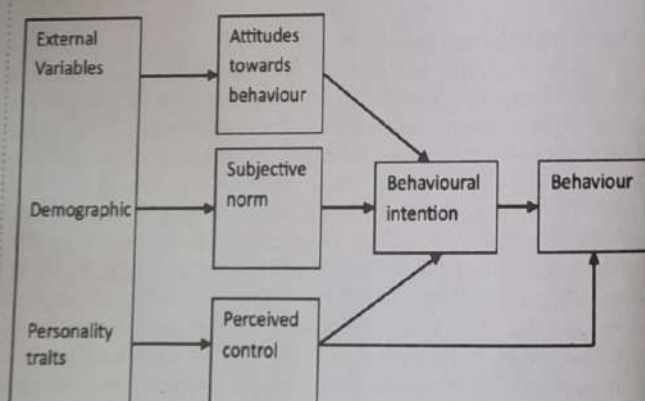


Fig: Theory of Planned Behaviour (Conner and Sparks, 1995)

Basic concepts in Theory of Planned Behaviour

Behavioural belief: an individual's belief about consequences of particular behavior. The concept is based on the subjective probability that the behavior will produce a given outcome.

Attitude toward behavior: an individual's positive or negative evaluation of self-performance of the particular behavior. The concept is the degree to which performance of the behaviour is positively or negatively valued. It is determined by the total set of accessible behavioral beliefs linking the behavior to various outcomes and other attributes.

Normative belief: an individual's perception about the particular behavior, which is influenced by the judgment of significant others (e.g., parents, spouse, friends, teachers).

Subjective norm: an individual's perception of social normative pressures, or relevant others' beliefs that he or she should or should not perform such behavior.

Perceived behavioral control: an individual's perceived ease or difficulty of performing the particular behavior (Ajzen, 1988). It is assumed that perceived behavioral control is determined by the total set of accessible control beliefs.

Control beliefs: an individual's beliefs about the presence of factors that may facilitate or impede performance of the behavior (Ajzen, 2001). The concept of perceived behavioral control is conceptually related to self-efficacy.

Behavioral intention: an indication of an individual's readiness to perform a given behavior. It is assumed to be immediate antecedent of behavior (Ajzen, 2002). It is based on attitude toward the behavior, subjective norm, and perceived behavioral control, with each predictor weighted for its importance in relation to the behavior and population of interest.

Behaviour: an individual's observable response in a given situation with respect to a given target. Ajzen said a behavior is a function of compatible intentions and perceptions of behavioral control in that perceived behavioral control is expected to moderate the effect of intention on behavior, such that a favorable intention produces the behavior only when perceived behavioral control is strong.

The Theory of Planned Behavior specifies the nature of the relations between beliefs and attitudes. According to these models, people's evaluations of, or attitudes toward behavior are determined by their accessible beliefs about the behavior, where a belief is defined as the subjective probability that the behavior will produce a certain outcome. Individuals' elaborative thoughts on subjective norms are perceptions on whether they are expected by their friends, family and the society to perform the recommended behaviour (Ajzen & Fishbein, 2005).

Application of the theory of planned behaviour to the study

The theory of planned behavior looks beyond the individual cognitive space and considers social influence such as social norms and normative belief and individual perception and locus of control. The theory also emphasizes the need to understand the reasons for the action. Example, various people's attitude and behaviour toward Covid-19 vaccination; a person can learn about the Covid-19 vaccination, believe that change is possible and through support from the environment will act on the information. The intent of the individual which is influenced by peers, the media and social norms to perform a specific behavior is based on this theory. All information is critically analyzed by the individual and decision made to participate or not. The greater the intention is, the more likely the actual behaviour.

Attitudes towards behaviour (Covid-19 Vaccination): The attitude is determined by the belief that undergoing Covid-19 vaccination will ensure prevention of the disease before incidence

becomes very high. This belief about the value and consequence of the vaccination will determine one's participation.

Subjective norms: Many people's attitude and behaviour is influenced by the value and opinion of significant people around them. Thus vaccination for Covid-19 infection has not been so popular among communities so there may not be adequate motivation of the people by significant persons. If approval from others is lacking, many may not be willing to participate. Thus a person's behaviour is highly influenced by the judgment of significant others (e.g., parents, spouse, friends, teachers and co-workers).

Perceived behavioural control: The individual's behaviour is determined by his or her locus of control. If internally determined the person will believe that his participation in the vaccination will reduce his risk of having Covid-19 infection but if externally determined, he will depend on fate to take its course and will not take preventive action. His belief also about access to the resources needed for successful vaccination would also determine his behaviour.

Socio-demographic variables: Various factors such as level of education, socioeconomic system, age etc all contribute in conditioning people's attitude towards Covid-19 vaccination, subjective norms and perceived behavioural control.

Behavioral intention: Various factors determine behavioral intention. For anyone to participate in any vaccination programme the person has to be ready. If he is not ready to be vaccinated, he will not present himself for the exercise. His behavioural intention is based on attitude toward the vaccination, his belief and that of people around him about the need for and the benefits of vaccination and his feeling about being in control.

Behaviour: This is the act of accepting to be vaccinated. His uptake of Covid-19 vaccination is a function of compatible intentions and perceptions of behavioral control. His intention to participate in the exercise will make him utilize available services. Counseling is a major factor that can improve the person's perceived behavioural control, as this will provide the necessary avenue to empower him with the relevant information that will enable him make informed choice. The desired change can be planned and achieved through counseling.

Example III: Role theory

Role theory is used in explanation of social behaviour, and it consists of loosely linked hypothesis and a set of rather broad concepts (Wrightman and Deaux, 1981). Role theory explains behaviour in terms of the roles, role expectation and demand, role set and reference group operating on the participants in the set of behaviours or function associated with particular position within particular social content (Wrightman and Deaux, 1981).

Roles have behavioural references concerned with standards of expected behaviour or judgement about behaviours. They reflect goals, values and sentiments operating in a given situation. A role involves an action of an individual in a particular setting with identified expectations as well as to their own feeling (Doheny, Cooper and Stopper, 1992). Role is independent of the holder.

Doheny et al (1992) stated that "the role elements differ from the nursing functions, functions can change from setting to setting, however, the elements are constant and utilised over and over". The elements of the nurses' role include that of a caregiver, client advocate, counsellor, educator, coordinator, collaborator, consultant and change agent. Every nurse as a professional should be able to perform the roles identified in dealing with issues relating to disease prevention. This has implication for knowledge as the person needs to be versed in the area before she can educate and counsel. The nurse as an educator may fail to perform her role due to lack of knowledge, anger and anxiety from a perceived lack of time

increased responsibility in carrying out this part of the role (Doheny et al, 1992).

According to Webber (1979) "each person in an organisation is expected by his or her superiors, peers, subordinates, others to behave in certain ways (called role demands). However, the individual involved may not perceive his or her role exactly as they do. The said role perception may be covert or may reflect wishful thinking, what one will like the role to be". This shows that even though role is defined according to position but individuals may not appreciate some of their roles.

The actual behaviour or role performance grows out of one's reconciliation of differing perception. Role effectiveness is determined by the accuracy of the person's understanding of the role demands and how well the behaviour meets the role demand, while role satisfaction depends on how closely behaviour fits the person's role desire (Webber, 1979). This can be achieved when the knowledge acquisition is commensurate with role demand. With good knowledge, the nurse perceives her role correctly and this results in correct processing of information. Information correctly processed with availability of resources will result in role performance and improved practice will be the output. This can be applied in cervical cancer screening.

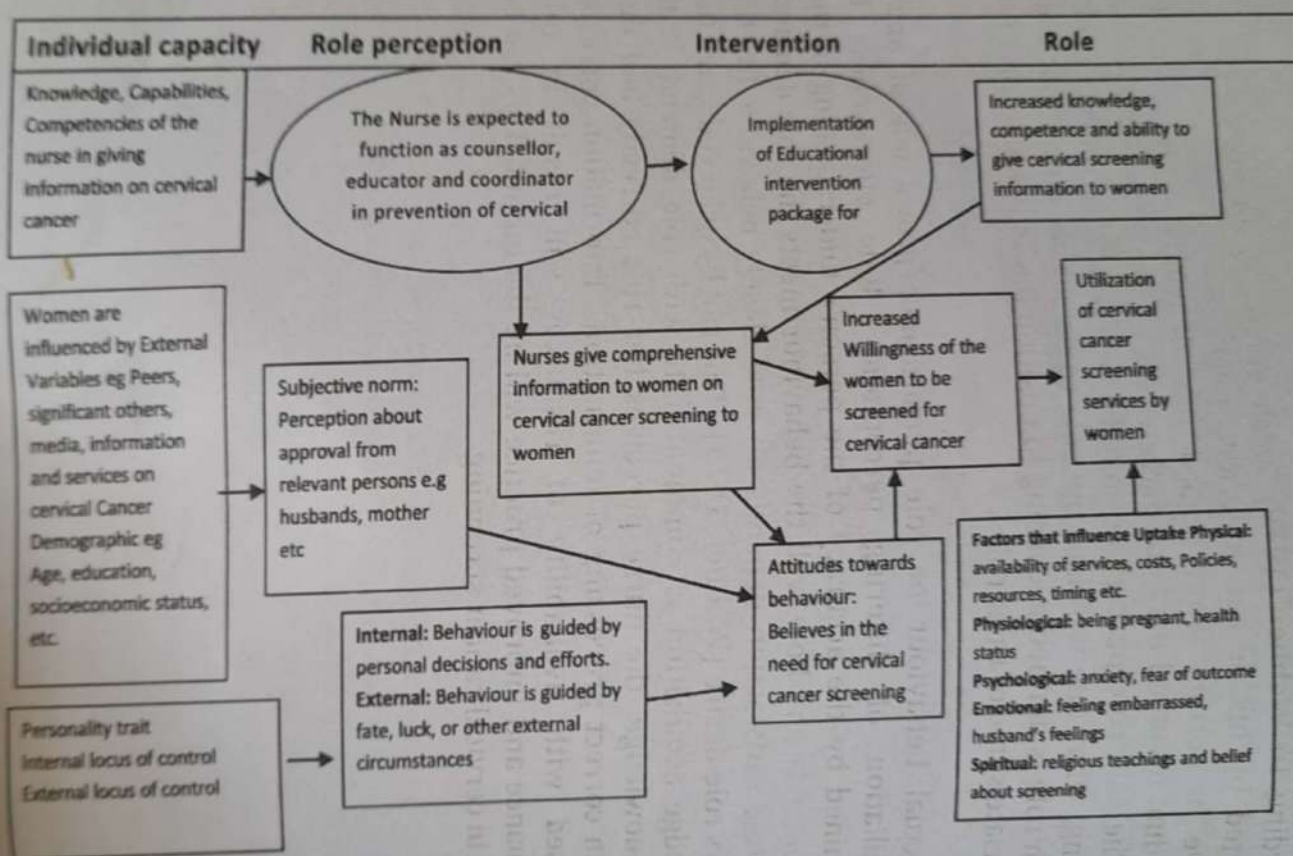


Figure 2.3: Conceptual Framework for Nursing Educational Intervention for Cervical Cancer Screening Services Utilization

Conceptual Framework

The conceptual framework for this study was derived from concepts from theories of planned behaviour and role theory. The conceptual framework tries explaining relationship between the variables central to this study on cervical cancer screening uptake.

The nurse has the important role of giving information to women and providing necessary care. The women as care and information recipient are influenced by various factors which the nurse needs to be aware of in order to ensure that the goal of preventing cervical cancer is met.

Individual capacity

The Nurses' Knowledge of cervical cancer, Personal Attributes- Capabilities, Competencies, commitment as well as previous experience and Educational qualification will influence their ability to perform their role in cervical cancer screening. On the other hand the role set being women are affected by various external, demographic and personality variables. All these go a long way in determining their role perception.

Role perception

This is influenced by the nurses' role demands. What she feels is expected from her in terms of cervical cancer screening will determine her effort in cervical cancer screening.

Role Demands: The Nurse is expected to function as a counsellor, educator, coordinator, collaborator, consultant and change agent in the prevention of cervical cancer.

Attitude towards behaviour: A woman's attitude towards screening may be negative or positive depending on the influencing factors like her perceived control over her health, approval from relevant persons like the spouse, the nurse, doctor etc and access barriers to cervical cancer screening services. The nurse plans intervention based on the areas of identified difficulties.

Intervention: These are measures instituted to achieve a desired goal. They are prescribed in a given situation to meet identified needs of the clients and health care providers. Educational intervention to ensure that behaviour needed to prevent cervical cancer is implemented. Structured educational information given to nurses to improve their knowledge and capacity. Also information on cervical cancer screening given to women by the nurses. If the intervention was effective then the woman will have a positive attitude towards uptake of cervical cancer screening which is demonstrated in her willingness to be screened and actual participation.

Role performance

Nurses give comprehensive information to women on cervical cancer. Increased knowledge, competence and ability to give cervical screening information to women motivate the nurses to provide necessary information. The regular provision of cervical cancer screening information by nurses in turn increases the willingness of the women to participate in screening. There are the factors that may influence actual Uptake of cervical cancer screening services. These need to be identified and addressed in every given situation. They could be:

- Physical factors like availability of services, costs, Policies, resources, timing etc. if the services are not accessible they may not be utilized.
- Physiological like being pregnant may make a woman want to postpone screening, general health status may also influence uptake negatively or positively.
- Psychological: anxiety, fear of outcome or result may influence final uptake.
- Emotional: feeling embarrassed, husband's feelings
- Spiritual: religious teachings and belief about screening may affect uptake

The difficulty could also arise from external variables like availability of services, social norm, lack of support from peers and significant persons. Also personal variables like her locus of control, level of education, age, parity, marital status may be contributory to attitude towards uptake of services.

Role effectiveness

This is determined by the nurse's ability to give information to women and women's acceptance of the information given on cervical cancer. The nurse's role is effective if the women develop positive behavioural intention. **Nurses' Knowledge** of cervical cancer, personal attributes and capabilities will influence their ability to perform their role effectively in cervical cancer screening. **Behaviour:** the women who believe that they can contribute to their health will likely utilize health promotion services. Thus information received will increase the women's willingness to utilize cervical cancer screening services. Thus, on the final analysis may influence their actual utilization of cervical cancer screening services.

Example IV: (Osuala E.O. Thesis: Community Health Nurse led intervention on prevention of Hypertension among rural dwellers in Isunjaba. Imo State. 2014, Ibadan).

The PRECEDE-PROCEED model: This Model uses deductive reasoning in applying an epidemiologic approach to community health planning. It was developed by Green and Kreuter in 1991 and has since been reviewed (Green, and Mercer, 2010). It is a health promotion planning framework consisting of two components: diagnostic components called PRECEDE and the implementation and evaluation component PROCEED. PRECEDE was conceptualized by the authors as an acronym for Predisposing, Reinforcing, and Enabling Constructs in Educational Diagnosis and Evaluation. PROCEED is an acronym for Policy, Regulatory, and Organizational Constructs in Educational and Environmental Development. The PRECEDE model can be used at the primordial and primary (health enhancement), secondary (early-detection), or tertiary (therapeutic) stage of prevention or treatment and may be viewed as an intervention whose purpose is to "short circuit illness or enhance quality of life through change or development of health related behaviour and conditions of living" (Green and Kreuter 1991).

The PRECEDE-PROCEED framework is used in the determination of the population needs prior to a program implementation, again the achievement of program objectives can be measured throughout the programmes phases as evaluation methods are established in the planning phase. The model also was specifically developed with the community in mind rather than individual making it well suited for community intervention. This makes it unique

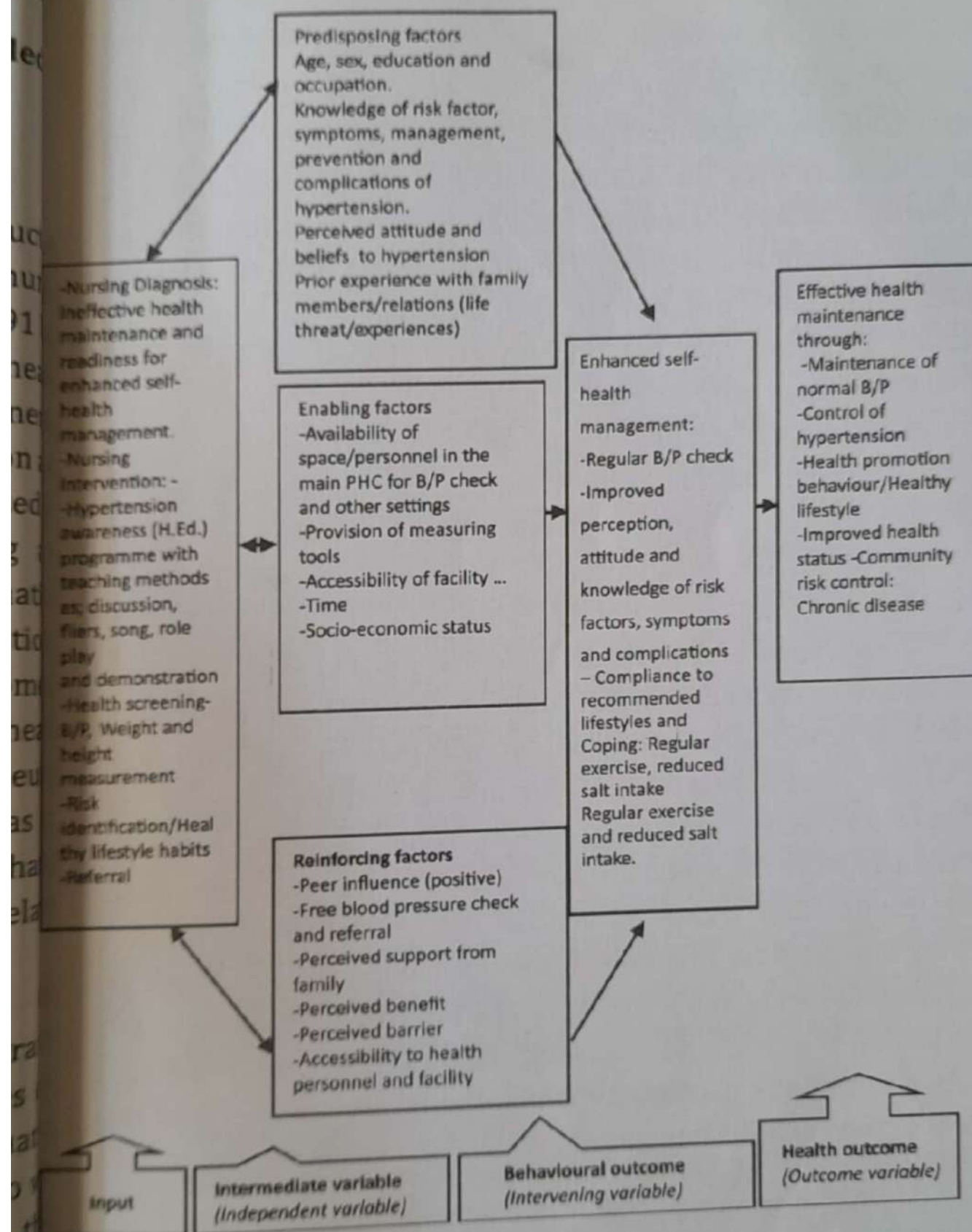


Fig : Application of precede model in the prevention and control of hypertension.
Adapted for Green and Mercer (2010) precede-proceed model.
Available at <http://dante.med-utorontoca/doch/precede.htm>.accessed 15/3/2010

PRECEDE consists of five steps or phases while proceed is composed of four additional phases. All the five steps in PRECEDE involve diagnoses in the following order: social, epidemiological, behavioural and environmental, educational and organizational and lastly administrative and policy. PROCEED follows with the 6th phase (implementation of programme and services). The interventions identified in phase five are implemented. Phase seven (process evaluation) entails process evaluation impact of those interventions. Phase eight (impact evaluation) involves evaluating the impact of the interventions on the factors supporting behaviours, and on behaviour itself. The ninth and last phase (outcome evaluation) comprises of outcome evaluation. In actual practice, PRECEDE and PROCEED function in a continuous cycle.

The precece model which is adopted for this work focuses on relationships among factors that could affect level of awareness and possible adoption of behaviours that will lead to prevention and control of hypertension. The factors are three namely, the predisposing factor, enabling factors, and reinforcing factors, referred to as the intermediate variables. The nursing intervention is the input. The input influences the intermediate variables resulting in behaviour outcome of regular blood pressure check and compliance to lifestyle modification. The intermediate variables; the predisposing factors, the enabling factors and the reinforcing factors, serve as the process by which the intervention would thus influence the health behaviour which is adoption of positive lifestyles to hypertension prevention.

The predisposing factors precede the adoption of positive lifestyles to hypertension prevention. The factors are age, level of education, sex, occupation, attitude and beliefs, life experiences as

well as knowledge of risk factors, symptoms, management, prevention and complications. Motivation for behaviour change is largely embedded in predisposing factors. Enabling factors include availability of manpower, money, facility, tools and time. Reinforcing factors are those factors subsequent to health behaviour that provide continuing reward or incentive for the behavioural outcome. The reinforcing factors include free services, perceived benefits, and barriers, accessibility of health facility and personnel, family support, and availability of measuring tool. The hypertension sensitization programme is expected to increase level of knowledge of risk factors, symptoms, management, prevention and complications of hypertension, correct erroneous perceptions and beliefs (myths), promote adoption of healthy lifestyle habits in relation to hypertension prevention, and reveal incidence of the disease. The health education package was dependent on level of knowledge, attitude, and lifestyles habits of the subjects in relation to the problem. It is therefore expected that the aforementioned intervening variables will collectively bring about the desired outcome. The input is expected to make an impact on the predisposing and enabling factors. For the evaluation of the intervention programme, the factors under the predisposing and enabling variables shall be measured as intermediate or independent variables that have led to adoption of regular blood pressure check and compliance with recommended lifestyle.

II. EMPIRICAL REVIEW:

Empirical review focuses on the related works done in the area of study/topic by various researchers: locally, regionally and internationally. This guides the study and provides relevant information for the discussion of the

work. On the other hand, theoretical review deals with documentations from books and authors of various articles on the topic or variables being investigated. Example: According to Osuala, (2014) "The prevalence of hypertension in Nigeria is higher amongst women" All the narrative about this statement belongs to empirical review while "the pathophysiology, symptoms or factors contributing to hypertension by Smeltzer et al. 2020" for example is termed theoretical review.

STUDY DESIGN

What is Study Design?

Designing is a term to describe a number of decisions which need to be taken regarding the collection of data before the data are collected. Study design simply means the style, type of study or method to be adopted in carrying out the study. For example, if one is investigating on the issue: 'induced abortion is on the increase in the last two weeks', as a researcher, this is a problem. One needs to prove this, find out why, investigate and compare results. The researcher has to decide on the type of study design to be adopted. The design chosen must be appropriate because if a wrong design is applied to a problem, it will be a wasted effort. In choosing a study design, certain issues must be considered. Ask yourself these questions.

- What is the objective of the study?
- What is the problem?
- What do I want to achieve or describe, is it the size of the problem or the nature?
- What are the reasons why certain things happen?

Study design depends on the objectives. Objectives could be in two forms; to describe the situation as it is. For example, the required

information may be needed for health planning or the information obtained may be necessary to provide intervention for the problem. One needs to have an idea of the problem in order to provide for its intervention. Objectives must be specific to deal with the problem. Extent of study design also depends on the resources available, which include capacity of the personnel, money and time available. This is important because the study design may be appropriate for a particular study but money may not be available. International agency's funding covers a wider scope than personal funding. Time frame should be considered because some international agencies might approve a study, which has to be carried out within a specific time frame. Time frame helps to determine the study design. The design can be *descriptive survey (non-experimental)* whereby the researcher describes the situation as it is or *experimental* whereby manipulation of the environment or subjects are involved. The non-experimental design can further have its various types attached to it for concrete signage of the particular design. Example; the descriptive study/design which is grouped under non experimental design/study may further be defined by noting the sub-types as cross sectional, quasi experimental, historical etc.

TYPES OF RESEARCH

Research can be classified by various approaches: function, design or content, depending on the author. The classification most commonly used is experimental and non-experimental. Hayes classified research as Basic Experimental and Non-experimental. French groups it as exploratory, descriptive and experimental. In this book it will be classified as Experimental, Non-experimental (descriptive) and Analytical

A. EXPERIMENTAL

This is conducted under highly controlled conditions. Experimental research is an approach in which laboratory or a field animal is used in tests or trials. Subjects are selected and

allocated randomly into either treatment or control group. The experimental group is exposed to the variable under study whereas the control group remains unaltered. The result of the tests made on the two is compared. Experimentation is the most powerful research approach yet designed. It is mainly used in the physical sciences but less popular than survey approach in health (nursing) research due to ethical consideration when applied to human beings. It requires approval from subjects, governing and review agencies. More preparation is needed and high level of creativity on the part of the researcher is demanded. The environment, subject or both are usually manipulated. The natural state of the subjects or environment are altered purposely by the researcher. In an experimental study, the investigator determines through a controlled process the exposure for each individual (clinical trial) or community (community trial), and then tracks the individuals or communities over time to detect the effects of the exposure. E.g. (new vaccine vs. placebo) to see whether the vaccinated group has a lower rate of disease. Similarly, in a trial to prevent onset of diabetes among high-risk individuals, investigators may randomly assign enrollees to one of three groups — placebo, an anti-diabetes drug, or lifestyle intervention to investigate impact on recipients of each group.

B. NON-EXPERIMENTAL

Non-Experimental study as the name goes does not involve field animal and laboratory experiment rather human beings are usually involved in the study.

Non-experimental include the following:

Descriptive/Survey

Historical

Case Study
Exploratory
Explanatory
Evaluatory
Aesthetic

Descriptive Study

This is also called survey by many authors. It involves description of events, situations and phenomena. It is done by studying persons, place and time. Certain questions guide a descriptive study: Who are they; children or adults, males or females? Where is it; Eastern Europe, South West of Nigeria? When is it; night, day, dry season? In a survey study (descriptive), a body of data is collected, recorded and analyzed. Survey study is a descriptive, non-experimental research, which is used to answer questions or gather information on a particular issue or event like: do girls in Aghamelum marry early? Or awareness and attitude to teenage pregnancy among students of Egbu Girls Secondary School Owerri, Imo State etc. It could be used to satisfy curiosity, e.g., nurses marry after 30 years of age. It can be used to solve problems, describe the disease pattern in communities among populations, or establish cause-effect relationship e.g. relationship between bottle-feeding and diarrhea. Any type of research that does not involve experimentation is categorized as survey by many authors. Others limit their definition of descriptive study or cross-sectional type of research where a large number of cases are investigated (Surveyed). The contact with subjects and collection of data is once and for all. Note that most nursing research come under this category of survey.

Historical Study

This is the integrative account of the relationship of people, places and events in a historical perspective. It looks into the past in order to explain the present. Nurses are using historical research to study the lives and contributions of important nursing leaders. An

example is the research that was carried out on the activities of Florence Nightingale to show the autonomy of nurses since she was autonomous during the Crimean War. Note: Disadvantage is that it can be subjective.

Case Study

This type of study focuses on a single subject and can be thought of as an in-depth investigation depending on when the subject lived. Its similarity with a survey or historical study lies in the fact that it is based on description. A person can be studied in the hospital as a *case* if the disease is rare or the symptoms and signs presented is new in health or not associated with already known pathophysiology of the diagnosed condition.

Aesthetic Study

It is also descriptive, but it is tailored towards research in music, church and in ornamental areas as art. It is done in appreciation of the beauty of something that is nice.

Evaluatory Study

Evaluatory research is done to find out if an introduced factor has worked or not. It is different from quasi experimental study as it does not involve control and experimental group; rather, an event, situation or course can be evaluated. A person's health condition or educational performance can be evaluated. Note: its usefulness lies in the basis for improvement.

Explanatory Study

Explains 'why' and 'what' of the findings eg. "Negative response of adolescence sexual experience". Using this research design, one finds out whether or not adolescents have negative or positive response to sexual experience as well as reason (s) for their response (s) Explanation is given for responses or actions.

Exploratory Study

This as the name indicates, examines an issue or situation thoroughly in order to test it or find out more things about it. For example: "Do students in Iyi-Enu go on family planning pills?" The researcher will not only find out whether or not the students use pills but will go further to explore the circumstances surrounding the use or non-use of it among such population. Reasons for actions or inactions (responses) are sought for.

C. ANALYTICAL STUDY

Analytical study can be experimental or non-experimental. It is usually employed if testing of hypotheses is involved. Analytical study is used when the determinant or cause of disease or situation is to be assessed. It answers question as to 'why' and 'how' the situation, event or disease occurs. There are three (3) types of analytical studies.

i Cross Sectional Study

This aims at testing of hypothesis, which suggests an association between population characteristic or agent and occurrence of a disease. It describes as well as examines factors associated with the occurrence of a disease. It also involves a single examination of a cross section of a population at a given point in time as the name applies. Note that when the testing of the hypothesis is related to two different points. (longitudinal study is involved). In cross sectional study, data is collected and analyzed in relation to the disease and its population characteristics such as age, sex, occupation, social class, educational level, e. t. c, of those involved in the study. Since data is collected at a given point in time, it yields information on the number of existing cases at a given point in time that the study took place. Therefore, they are sometimes referred to as prevalent studies. It differs slightly from descriptive study, in that actual subjects must be studied as opposed to the use

of routinely available data, which may be employed in descriptive research design.

Similarities between cross sectional and descriptive studies

- Similarity is that both are cheap, easy and quick to conduct. Both involve a single examination of a cross section of the population at a given point in time.

Differences between cross sectional and descriptive studies

- Cross sectional study describes and also examines factors associated with the occurrence of a disease i.e. It also examines the relationship between a population characteristics such as age, sex, etc. and the occurrence of the diseases. Descriptive study simply describes the situation. Cross sectional study is population based while descriptive study may depend on routinely available statistics for its analysis.

Disadvantages of Cross Sectional Study

Since data on population characteristics or causative factor is being investigated at the same time together with the disease pattern, it is difficult for the investigator to say with degree of certainty, which came first, the disease or the causative factor. Results may therefore sometimes be very difficult to interpret. Example, in a cross sectional study on association between lung cancer and smoking, the degree of certainty as to whether smoking resulted to lung cancer or if smoking was commenced after onset of lung cancer may be difficult to establish.

ii. Case Control Study

Case control study is an observational, non experimental design, aimed at testing a hypothesis on the association between a suspected etiological agent and occurrence of a disease. It compares those with history of exposure to the disease (cases) with those

without the disease (controls). Cases are those with particular disease while controls are those without the disease. In both groups, exposure history of the mentioned etiological agent is looked out for. The investigator notes the previous history of the cases; finds if they were previously exposed to etiological suspected factor. This is why case control study is also referred to as retrospective study. Retrospective delves into the past while prospective examines future occurrences or events. An example of this type of study is on cigarette smoking and pulmonary disease where there is a relationship or association between the disease and the habit. In a case-control study, investigators start by enrolling a group of people with disease. As a comparison group, the investigator then enrolls a group of people without disease (controls). Investigators then compare previous exposures between the two groups. The control group provides an estimate of the baseline or expected amount of exposure in that population. If the amount of exposure among the case group is substantially higher than the amount you would expect based on the control group, then illness is said to be associated with that exposure.

Stages of Case Control Study

- The objective is stated and clearly defined. The objective is examined for association between group A. (case) and group B (control) and suspected etiological agent.
- The study population is defined, as the result of the study is expected to be extrapolated from the study population. Everybody in the community is included in the study but the larger population is eligible for the study. For example, examining a particular disease among women of the Christian Women Organization (C.W.O) Awka, eligibility will include all members of the union, whereas the target population will involve every member in Nigeria. The study population can be defined further as every member in Awka

that is less than 45 years of age. Note that study population different from target population.

- c. Cases are properly defined. This is important because many other factors depend on the definition of the cases. e. When studying women with cancer of the cervix, cases may be defined based on the investigation carried out in laboratory. Since cancer of the cervix has so many stages, one should decide at what stage of cancer, cases should be drawn. Some may choose to use cases in the terminal stage or second stage as informed by the objective or the researcher. Case control study can be used to assess the compliance or effect of a drug. Those on whom the drug is applied are the cases while non-compliers are the controls.
- d. Stage four involves selection of cases and controls. The basic principle of selection is that both cases and control must have common background/characteristics for a conclusion to be drawn. The cases should be similar to controls except in the occurrence of the disease; they must come from the same source for acceptability of the result/outcome.

Source of Cases

The Source may be from hospital, where there are already patients with the disease. The researcher states clearly who the cases are, inpatients or out-patients? Study population must be defined. Since source is the hospital patient, it means that everybody included in the study must have equal chance of getting the disease. It is assumed that all the cases come to the hospital, for example tuberculosis. Selection need to be made from a defined population in the disease being studied is such that the chances of the patients attending the same hospital is minimal, then the appropriate place to select from would then be the general population.

Source of Control

Ideally they come from the same sources as cases to ensure that one is comparing two entities that are alike. In the example with the hospital patients, control should be similar with cases except that cases have the disease while controls do not. The chance of utilizing hospital facilities is same for cases and controls. Selection can also be made from the general population. The control should have everything in common with the cases except that controls do not have the disease. Another example is the provision of school meal to pupils in a selected school (case) and non provision for pupils in the neighbouring school (control) in the same community. At the end of the feeding plan, nutritional status of the pupils in both schools would be assessed to note if there was any difference in the case and control.

- e. Matching Cases and Controls: The last stage of case control study is *matching*. The subjects for the study are then matched. When matching cases and controls, one must take care of confounding variables. The confounding variables are factors that are related to both occurrence of diseases and the causal or the risk factor under observation. Matching helps to take care of attributes which influence exposure and disease occurrence, example is age, gender, occupation etc. Assuming in a study on the use of contraceptives and ovarian cancer was conducted; and the study group are people with ovarian cancer and those with no history of ovarian cancer but the two groups have something in common i.e. use of contraceptive. At the end of the study assuming one ended up with the result as follows:

Ovarian cancer (contraceptive used)	No ovarian cancer (contraceptive used)
40%, mostly young people	30%, mostly elderly people.

The above result implies that prevalence of ovarian cancer is lower in the elderly on contraceptives. This may not be the true picture. One can only conclude and accept the findings if the confounding variables were taken care of before comparison of the two categories was made. The above result shows many younger people with ovarian cancer therefore it is not matched properly. The example given shows that there is a relationship between age and exposure. This can be taken care of by making cases 20-40 years and controls also 20-40 years. By so doing, the result will be acceptable. Ideal situation is to interview case/control but where the individual is dead, his death certificate can be used. If the study is on the influence of educational level on disease occurrence, the respondents' case notes can also be used.

iii. **Cohort Study (Longitudinal):** It is a non-experimental or observational study that tests or investigates possible relationship between a hypothetical etiological factor in persons exposed or unexposed to the factor and the disease occurrence. In cohort study one starts with the history of healthy individuals and goes on to find out who is exposed or not exposed to the substance under study. Cohort study can also be referred to as prospective study. Cohort means group of persons who share a common experience within a defined period of time, example marriage, birth, and occupational cohorts; such groups are followed up in time for development of the disease or the problem. The study group may be classified based on the degree of exposure to the suspected etiological agent or risk factor of the disease or condition under study. Example: workers at Nigercem company Enugu can be grouped and then followed up to establish the fact that they run the risk of pneumoconiosis as an occupational hazard, according to years of exposure in the

industry. Subjects are observed from time to time in view of noting certain characteristics under study. Jean Piaget's work on cognitive development is an example of a theoretical framework on which one can base a study of this nature (psychological development). Cohort studies start with a group of exposed and a group of unexposed individuals. These groups are then followed up over time and assessed to see who develops the disease. The incidence rate of disease in the exposed group is then compared to that in the unexposed group, thereby measuring relative risk (RR). Since we only select subjects by exposure, we can study several health outcomes at the same time. By properly selecting groups of exposed people we can study relatively rare exposures. Detailed information on confounding factors can be collected and controlled either through the analysis or design. The disadvantage of cohort studies is that they can take long time and generally are expensive. For rare diseases, the number of subjects that need to be studied is often quite large (majority may likely not show the disease), making cohort study impractical or unfeasible.

CHAPTER 2

ETHICS IN RESEARCH

Ethics: This is a branch of philosophy that deals with moral principles that govern or influence a person's behavior or action. In research one is concerned with the ethical and moral issues involved in gathering data on individuals. Information gathered on individuals on health related matters are held confidential unlike the mass media where freedom to publish all manner of scandal about the general public is possible, e.g. Reports on divorces, arrests, trials and frauds are given exposure to the world, even though publishing these proceedings cause a great deal of anguish to the persons and families involved, such effects are not considered by the press. Psychological tests are sometimes too probing and too personal, prying into lives of the respondents. In the fields of medicine, health and nursing, researching on the behavior and activities of people are involved; the issue becomes more critical when it involves the manipulation of human beings as they are placed in a laboratory experiment, either knowingly or unknowingly. As life has become so duplicate, ethical consideration is of paramount importance in health related studies to avoid litigation.

Objection to manipulation

Many scholars object to manipulations in research for several reasons. Information collected could be used for blackmail or extortion. Secondly, the motive for such a study may be questionable. These are the main reasons for objection by Nicholas

Von Hoffmann. Others like Horowitz and Rainwater did not agree entirely with Von Hoffmann. They are of the view that news media and other journalistic endeavors use opinions, attitudes and interviews as sources of information in same manner as other researchers. Sociologists usually study or observe their subjects and do not change or manipulate the lives of the subjects. However, they do agree that some sociological findings can become political statements. Philip Derek on his own contribution says that the material and knowledge gathered during research is not enough to allow governmental agencies to manipulate or otherwise control the lives of their subjects. Even though many benefits arise from research; nevertheless some ethical considerations need to be applied in the study of human beings. Social scientists conduct research for prestige and future studies.

Research in pure sciences produce results to save lives and improve health, while gaining status to the investigator. The ethical considerations may therefore be on the aspect of the investigator's freedom to present to a wide variety of readers, materials gathered on unsuspecting or even informed respondents. Some professional and technical research from pure sciences (Chemistry, Physics) can be published in their entirety since the results are as beneficial as the technique used in arriving at the findings. In the health field, generally all types of research findings should be open to publication though some people feel that revealing their personal health deficiencies and illnesses is really harmful to them personally. Publishing the findings is advantageous in the long run to the medical profession and the public at large. Obviously, this freedom should not be abused. One should therefore consider what

is ethically and morally acceptable in gathering or reporting information about human subjects.

Consequences of Manipulation

Any research in health field, like medicine or nursing can be used for good or evil. Example, if it is found out that patients recover better with individualized care i.e. concern for the patient as a person than grouped care which is concerned with task and disease, it would be immoral not to publish the findings simply because we are afraid that a nurse will resort to group nursing care if the patients are many. In medical practice, there is reduced freedom in studying the human body. In the past, dissection of cadavers was prohibited. Some countries only allowed a single dissection every five years. In order to appreciate the consequences of manipulations and ethical consideration in research, some scholars, Reynolds and Paul D. classified consequences of research into five categories, namely:

- i. **No Effect:** Here there is neither positive nor negative effect on the subject participating in the research. Most of the surveys for course requirement fall into this category.
- ii. **Temporary Discomfort:** In this category, the participant experiences temporary anxiety, tension, or physical pain. The discomfort terminates at the end of the study and is not more than what is encountered in a day-to-day activity; example, a student when asked to give a brief history of his life unexpectedly will experience temporary discomfort as evidenced by sweating or palpitation.
- iii. **Unusual Level of Temporary Discomfort:** This may last beyond the study. The anxiety, tension or physical pain may not

terminate immediately after the study. Post experimental treatment otherwise known as debriefing may have to be given, whereby the subject is told exactly what the study is all about, or why some part of the study has to be kept secret. The subject should be made to understand that the secrecy is not to trick him but to avoid bias.

Risk of Permanent Damage: This type of damage can be seen in medical field but not social researches. Under this category, whatever damage that is caused to the subject may be irreversible.

Certainty of Permanent Damage: This category is rare. In the health sector, lower animals are used for studies that are presumed to bring about permanent damage. Nursing or social research does not involve any permanent damage to the subject unlike some journalists, who through the use of cartoons can inflict or damage the image of individuals that hold top offices permanently. Health based researches do not ridicule anybody by name. Ethically, the study should pose no adverse effect or temporary discomfort on the subjects. Ideally, on no account should a study bring a permanent damage to the subject(s).

Guideline for Ethics in Research.

Certain steps have been adopted nationally and intentionally to ensure the rights of subjects during a study. The need for free and informed consent of subjects is stressed. Every effort must be made so that the subjects understand the nature and purpose of the research. Confidentiality, is to be maintained and priority given to the rights and concerns of the subjects. Concern for ethics in nursing research has been evidenced on the national and international levels. Considering ethical problems involved in using human subjects, to evolve a good research, information should be gathered

in as natural a setting as possible. After the conclusion of the research, if privacy has been invaded or ethical issues are raised, the material should be published only after the subject has given his permission. A researcher should never betray the confidence of the subjects.

Closely related to the issue of moral and ethics is the question of methodology. If subjects are told what the researcher is doing and their permission gained, will their typical behavior change? When the researcher must use any or all means available to collect information, yet there are certain limitations when it comes to investigating human beings. The researcher needs to ask himself the following questions:

1. What risk would be involved to the patient? Will it be detrimental to the family or the subject?
2. Does the subject have the right to know the purpose of the research? The subject should have as much information as possible and will not bias the findings of the study. This is because some subjects can react, in order to impress, fool or please the researcher.
3. How much of the study situation or the research instrument should the subject know? For example, if you want to find out the level of awareness, how much do you tell the client about the tools to be used? The question of bias will guide the researcher in this issue. Always tell the subject the instrument that will help you arrive at the purpose of the study. Ask yourself, what effects are likely to arise in the subject and those around him during and after or following the research process.

4. Will there be Hawthorne effect? Mayor did a study in Hawthorne Plant in 1945, using plant workers. The study was aimed at finding out how a certain procedure leads to higher productivity. The instrument, gadget and attention given to the workers, led to increase in productivity. The increase in production of commodity in the company was not due to the method employed but due to the motivation.
5. Must the confidence of the subject be respected? Yes. The primary responsibility of the researcher is to his subjects. At all times, whenever possible, maintain anonymity, if names are supplied, use it all alone. Try and gain the confidence of the subject.
6. What type of relationship should exist between the researcher and the subject? What we should be interested in, is the type of respect the subject obtains from the researcher. It is associated with subjects' confidence in you as a researcher. If you have any information that will hurt the subject, you do not use it. If you must use it, make sure it is not connected to his name.
7. Do you have to show the result of your findings to the subject? In the questionnaire or at the end, ask the subject if he will be interested in knowing the result of the study. In that case, do not give them the method or analysis but send them summary of the whole work.
8. Does the research design involve any form of invasion of privacy? Ask yourself if you are prying into their privacy. The privacy of the subject should be respected. There are certain things that may not be in maintenance of privacy e.g. asking

for last menstrual period of a pregnant woman is appropriate since it will be needed in the calculation of her expected date of delivery.

9. Debriefing: The subjects should be informed ahead of time what is expected of them during the research. After, the process, they should be told that the research is over so that they will not continue with whatever behaviour they were asked to adopt when the research was in progress.

PLAGIARISM

Related to ethics in research is plagiarism. It is unethical to copy another person's work without referencing or acknowledging the author. Plagiarism means taking somebody else's words or ideas and using them as if they were one's own. It is a breach of copyright. It can also stand for an illegal copying of one's work. Quoting somebody's work or write up without referencing, quoting or acknowledging the individual is unethical. Plagiarism is almost similar to piracy though piracy usually involves "robbing" i.e., copying one's tape, book etc. for making money.

CHAPTER 3

RESEARCH PROPOSAL

Research, being a planned and systematic activity, some steps have to be taken before launching into actual study. The actual steps to be taken are usually described in writing as a document, which serve as the plan for guiding the total research activity. This written plan is called "Research Proposal". In a proposal, one states the problem. It is a preposition made for any research especially the one that requires evaluation or money from an organization. It is also used as a criterion for consideration of an applicant for admission into a higher degree programme. Proposal has to be written and given to the person or organization. In the proposal, the problem is stated; objective, the need for the study and why the individual must conduct the study is also stated. A short review of literature is given showing that there is need for the research. The methodology is stated i.e. Whether descriptive, experimental, survey, etc. how the data will be analyzed and findings reported, is stated. Is it in a book, journal, media, etc.? Estimation of the cost of the project i.e. budgeting is made and presented. Are vehicles, typewriters, personnel etc. going to be involved? Research proposal entails description of the processes the researcher will undertake.

Research Process

The process of research involves twelve major steps:

1. Formulate the problem
2. Review the literature

3. Formulate the framework
4. Formulate hypothesis
5. Define variables
6. Determine how variables will be quantified
7. Determine the research design
8. Delineate the target population and sample size determination (*Describe in detail*)
9. Select and develop method of collecting data
10. Formulate method of analyzing data
11. Determine how result will be interpreted/ generalized
12. Determine method of communicating/ disseminating result.

Many a time, some researchers especially beginners, start off with what they think is a researchable topic, move on to data collection with the hope that at the end of the exercise, they will come up with a research study, only to end up wasting time, energy, money and materials. The exercise ends up as a failure simply because it lacked a sense of direction. To avoid this mistake, like everything done in life, one must plan according to one's means. The researcher must choose a topic within his or her scope and plan systematically how to reach the end, ensuring that the desired goal is achieved.

The first step in research process is the selection of a suitable research topic. For most beginners, this is the most difficult and time consuming period. Some are very much perplexed. Many of them

exhibit research phobia for which beginners are known. A research topic commences with discovering a problem in one's field of study, whose solution will depend on the answers derived from relevant study. At times one may not try to discover a problem, topic may be derived from complaint by some group of people or individuals. For example, the issue of low male enrolment in schools in Anambra State, one may want to know why. Another is low patronage of patients/clients in a particular hospital or setting. Topics may be derived through literature search; online, published write-ups, journals and magazines. One may even take up from one's findings; a further research on the issue may need to be conducted. A researcher may find out that the incident of malnutrition is high among pregnant women in a certain locality and after going through such work, may want to find out factors related to malnutrition or its effects on pregnant mothers in the said locality. Local government, state, national and international bodies (especially non-governmental agencies) may sponsor a researcher in conducting a survey where the organization has a vested interest in the topic. Research topic may emanate from one's observation.

After identification of topic, the next step is to reframe the topic in form of a research question. The answer then provides the basis for addressing the problem. For example has malnutrition any effect on the pregnant mothers in the said community. Why does St. Mary's hospital have low patronage of patients? The researcher must make sure that the issue is topical; something current and in vogue. It should be relevant, important and feasible. It should also be practicable and realistic in relation to time and money available. The population must not be too large to contend with. Research proposal is stated as follows:

STATEMENT OF PROBLEM

The first step is to select a topic on your own though your lecturer/supervisor may influence the supervisee choice. One can also choose a topic out of curiosity or be influenced by the financial support for such research. It could be from literature review. The stimulus for pursuing a study can be provided by learning what other researchers have found as well as their appraisal of further research that need to be done. Hypotheses that were tested may spring up new problems. It may be topics or investigations from colleagues. Researcher's background and experience influence his choice of topic. A doctor or a nurse will likely choose a topic related to health issues, while a sociologist will choose from the field of sociology. In developing a research problem, we must realize that it is a difficult task. Researchers must be original and creative in their thoughts. The environment must permit freedom of enquiry. A good researcher must be inquisitive. A problem is not a problem until you see it as a problem. For those who do not see it as a problem, they will not see the necessity for a research in the area. A problem has to be exposed/ dissected and then reduced from the main problem to sub-problem that is researchable e.g. 'obesity causes hypertension'. This can be narrowed down to a particular gender e.g. "obese women are hypertensive"; then what group of women? "Pregnant obese women". It can go on and on and come down to "obese primigravida", "elderly obese primigravida". All the "coverings" of the problem has to be removed so that it can be seen clearly. If the research problem is too extensive, the researcher becomes bored or may regret starting the work at all. The problem must be defined/ explicit from the onset. In a research problem, we have to look at 'what and 'why' of the problem. The "what" is the problem you have

selected from the big problem? The "why" is the purpose for which you are interested in the topic?

Another source is based on theory and model; to test the theory, to see if it really works, or to apply it to some new situations among different study population or settings. Using categorization, topics could be drawn from areas of nursing education, administration and practice. Topics in nursing practice including activities related to provision of direct and indirect care to the patient.

Characteristics of a Research Problem:

A research problem should represent a situation in need of solution, improvement, modification or change. Ideally, it should have at least two variables. I.e. stating a relationship between two ideas e.g. Teachers marry earlier than nurses. The variables must be clearly and unquestionably stated. Nobody should confuse the nurse with the teacher. The problem must be as simple as possible. It must be within the ability of the researcher. The problem should be one that can be stated in declarative form; in a form of question or hypothesis.

Guideline for Assessment of Researchability of a Problem

Certain guidelines help one to decide whether it would be worthwhile to develop a problem into a formal research project. By asking certain questions about a problem; one can deduce whether the topic is researchable or not e.g. Is it feasible to conduct research on the problem? Appropriate tools or methodology may not be available, even time and subjects may pose a problem. Example, "effect of using certain procedures on patients with rare conditions". It might take years to accumulate a sufficient number of cases to yield valid findings. What of the health of the individuals; will there

be harm from the study? What of cost? Is the problem of sufficient importance as to take all the trouble concerned with research? Research involves money, time and effort. Does the problem involve much expenses? Is the finding worth these expenses? Is the occurrence frequent? What is the possible effect of the research findings on the health of the people?

To what extent can the finding of the study or the problem be generalized? At times research is evaluated according to the extent to which its findings can be applied to situations other than the research setting in which it was conducted. Is the finding going to improve the welfare of the entire populace? What need will solution of the research problem meet? If the purpose cannot be defined, the problem is clearly not researchable. The purpose should meet a real existing need.

Is the problem of interest to the researcher? Interest in the problem can serve as a strong motivation for the researcher to pursue his research project to its valid/logical conclusion and with intellectual honesty. There is a burning interest in his study that he forsakes everything else to spend all his time on his research. Does the researcher have competence to pursue the research problem? Has he sufficient knowledge in areas involved in the study to make appropriate decisions? Does he need consultants, psychoanalysts, statisticians, full time members or part time members to form a research team? These are the questions the researcher needs to answer to pick a good research topic.

AIMS AND OBJECTIVES

Here the purpose of study is stated. The aims are stated first in general terms and then as specific measurable objectives. The research questions, which study has to answer, will be set out. For

example, if a study is being carried out on hepatitis in pregnancy within a community, the aim may be stated thus: to identify the predisposing factors of hepatitis among pregnant women in that community or effective methods of prevention and control within the last two years. Like every objective, the acronym 'SMART' should be used to test the quality of the stated objective. The objective has to be specific, measurable, achievable, realistic and time bound.

The specific objectives for the above stated topic may read thus:

1. To know the incidence of hepatitis among pregnant women in that community.
2. To determine their level of knowledge of hepatitis.
3. To identify factors associated with its occurrence.
4. To identify ways of prevention and control of hepatitis among the group.

The rationale/relevance for the study is stated. The reason for the study will be stated. Why does the researcher want to get involved in the study? Of what relevance is the study to the researcher and the community at large? Thereafter, comes the research questions. The researcher will likely formulate questions that will throw some light on the issues being verified and also evolve probable solution to the problem, bearing ethical consideration in mind. The research questions should also go in line with the above four set objectives. For the aforementioned topic, the research question will go thus: 'What is the incidence of hepatitis among pregnant women in the particular community? Etc. Another example of research question on a topic assessing the knowledge, attitude and practice of reproductive health issues among adolescents can read thus:

1. What is the respondents' level of knowledge of physiology of female reproductive system?
2. Do they know the hazards of unwanted pregnancy?
3. Do they have knowledge of contraceptives?
4. Do they involve in sexual activity?
5. Do they use any contraceptive?

REVIEW OF LITERATURE

Three forms of review may be demanded by some establishments namely theoretical, empirical and conceptual as earlier mentioned in previous chapter. In research study, review of literature is an extensive and systematic examination of publications relevant to a particular research topic. It gives a brief description of some published works related to the one under consideration. The place where the study was conducted is mentioned; methodology and findings of the researchers are highlighted. The difference or relationship which the proposed study would address has to be discussed. This is referred to as empirical review.

Types of Review

Two types of review are described here:

- a. General Review
- b. Critical Review

General Review

In this type of review, all relevant publications are located and examined. It has to be in relation to the topic under research.

Critical Review:

Here all relevant publications are located with the view of identifying their strengths and weaknesses. The publication has to be of significant value to the particular study.

Reasons for Literature Review:

It reveals the extent to which theory and research work has been carried out in the field of study. It justifies the need for further study in the area or topic. The extent and quality of information gathered shows how conversant the researcher is in his field of study. It reveals any difficulties encountered by the previous researchers so that the researcher knows his or her limitations. In addition to these benefits it forms a base for the research work. A good review of literature will help the researcher to find a well-structured study that may serve as a guide to the present study. One can pick a researchable topic by going through other people's write-ups. It lends strength to the work as the discussion of findings has to fall back on the literature reviewed.

Steps for Literature Review.

There are two ways of searching for literature:

- Manual Search
- Computer Search

Manual Search:

The index in the library is used to locate the book or manual. If one is looking for ideas related to sociology, he should look for sociological abstracts in the index. The index may be subject or author based, but it has to be arranged in alphabetical order to identify the journal, the author, title, place and date of publication. In nursing research, cumulative index include International Nursing Review Index and Medicus. It is advisable to record the author, title

and year of publication etc. in a note book for future referencing at the end of the study.

Computer Search

The computer does the search. One can either conduct the search by himself if computer literate; otherwise he supplies the librarian/computer scientist with his topic. The librarian/computer scientist then feeds the information into the computer and the system produces journal that are related to the topic. Best articles are usually from the Internet. Books are also good, though the ideas contained therein, usually gets obsolete by the time the book is printed. Topical issues are usually found in journals, magazines, manuscripts and the like. With Google search or preferred search engine as PubMed, keying in the author(s), title or topic or the variables under study together with the desired year of publication, relevant data could be rolled out. More recent works/studies are preferable. Journal articles dated five years and below and books dated 10 years and below are recommended by many institutions. Most schools subscribe to such database as Hinari, Embase, Research 4 Life etc to make access to relevant data online easy for the students.

Organization of Information

The information extracted from various write ups are arranged under sections or sub-headings. The subheadings are guided by the objectives and variables under study. If the required information is on the knowledge of a certain situation, topic or subjects, findings of various researchers on the knowledge of the concerned group or participants on the issue should be taken together in paragraphs. It is then followed by information on any other issue like, attitude or

practice. The information must be stated in an unambiguous manner and must be related to the topic under study. It should be arranged attractively and orderly or logically. All authors mentioned in the text must be cited as approved by the institution or organization as reference format differ from setting to setting.

METHODOLOGY AND PLAN FOR DATA ANALYSIS

This is where the researcher describes the study design for the proposed study.

The geographical location, name of the area/setting, the population involved has to be stated. The sample size and sampling technique must also be stated. The instrument for data collection, method of administration and collection is described; method of data analysis is finally stated as well as the type of presentation. Ethical approval is obtained from an ethical approval committee based on proposal for the study.

SCHEDULE OR TIME/WORK PLAN

A timetable is drawn to show the plan of work. This usually forms a guide as circumstances beyond one's control may affect the schedule.

Here is an example of such timetable.

TASK	DURATION			
	JAN	FEB	MARCH	APRIL
a. Obtaining Permission	✓			
b. Training of interviewers		✓		
c. Pilot test		✓		
d. Field work		✓	✓	
e. Data Analysis			✓	

f. Reporting writing				✓
TOTAL	FOUR MONTHS			

BUDGET

Budget is an estimate or plan of the money available to somebody and how it will be spent over a period of time. When one wants to carry out a research, the available resources have to be considered. Most of the governmental and non-governmental agencies calling for proposals usually state the limits of their funding. The duration of the task, manpower and material involved has to be outlined in order to make a detailed and realistic budget. When budgeting, one should consider the travels that will be made and the mode of transportation of such travels for realistic costing. Consumables and equipment should be put under consideration. The people that will assist the researcher will receive their honorarium. This has to be put into account. Explanation on why the various amounts should be spent on various items has to be specified. For instance, one may have to justify the need for a jeep in carrying out an assignment, by stating that our roads are in a terrible shape.

Below is an illustration of a budget for a research (N420 = 1US Dollar)

RESEARCH BUDGET

REQUIREMENTS	NAIRA (N)	DOLLAR (\$)
Honorarium for 3 researchers @ N200,000 each	200,000	142.86
Salaries of 3 research assistants @	12,000	28.57

N5,000 per month (training Period + field trip)		
Hotel & feeding for all researchers during field trip @ N4,000 x 6x4	480,000	114.29
Transportation cost to 3 states for data collection @ N2,000 per month for 5 months, (N2,000 x 6 x 5)	150,000	35.71
Production of research instrument	100,000	23.81
Communication (Telephone etc.)	30,000	7.14
Library services	40,000	9.52
Stationeries for field work (Paper, pencil, biro glue, e.tc)	40,000	9.52
Photocopy of relevant materials	40,000	9.52
Computer Analysis	60,000	14.28
Production of final report & copies for dissertation	30,000	71.4
Contingencies	10,000	2.38
Total	1,620,000	404.74

REFERENCING

Any author that has been referred to is usually cited accordingly in the proposal, body of work and at the end of the report. The three main styles of referencing are the Vancouver, the Harvard and APA, (American Psychology Association) format. Citing author(s) in the body of the work (text) is referred to as citation while at the end of the whole documentation/write up or report, it is known as referencing.

a. Harvard System

In this type of system, the surname of the author is written in the text. At the end of the text, the reference is arranged thus: Surname,

textbook, edition, publisher, town and year. Page (Optional) i.e. Okeke B. U: Human Biology for Midwives, 2nd ed., Ochanja printing press. Ogidi, Nigeria. 2001. In the Harvard system, references are cited in the text using name of author (s) and date of publication, e.g. Malaria is one of the major causes of anaemia in children in Nigeria (Osuala & Ohiaeri, 2005). At the end of the work, all the references cited are listed in alphabetical order. The article cited above will appear as: Osuala, H. and Ohiaeri, D. (2005). Factors associated with occurrence of Anaemia in children in Anambra State, Nigeria. Family Health Journal. 24, 1: 12-16.

Note: For Journal/Articles, the sequence is thus: Surname, initials, (year) and 'Title', name of journal, volume/issue: month, page. I.e. Adeniyi Funke S. (2001). Patient Teaching In Maternity Wards, Nursing Today 4, 9: 20-23. If it is a book, Author (s) name (s), date of publication, title, publisher's name, place of publication and pages are stated. E.g. Okafor, C. (1998). An Introduction to Sociology. Lagos, Nigeria, 5-12.

b. Vancouver System

With this style, the authors cited in the text are numbered serially using superscript style. On the reference page at the end of the paper, the surname of the authors cited in the text is not arranged alphabetically as in Harvard style of referencing rather it is according to the numbering in the text. Arrangement of the referencing is based on the numbering which is done consecutively in order mentioned in the text. In Harvard style of referencing; only authors and year of publication is written in the citation and no numbering is done. Citation number is placed in the text after the name of author or statement in Vancouver style of referencing. E.g. According to Ukonu, (2020) Fertility age in African women is from

13 - 49¹. If no name is mentioned, the citation is placed at the end of the statement, e.g. it has been shown in Africa that malaria is endemic². On the reference page, full information on this numbers 1 and 2 authors would be stated accordingly. A maximum of three authors should be quoted, but if there are more, only the first three should be listed, followed by et al. Journal abbreviations should conform to Index Medicus National Library of Medicine approval. The arrangement of the referencing should be as follows: Surname of author (s) name (s) or initials, title of article, name of journal, year of publication. The Volume of journal and then the first and last pages of the article. This applies to the reference page only as the citation in the text bears only surname of author and year of publication as the case may be with the appropriate subscribed number attached.

Example, the reference that comes at the end of the study should be stated thus:

For Journal/Article

Surname, initials, 'title', Name of journal, year, (vol); number and page. E.g. (Amazigo U., Nancy S., Kaufman J. et al. Sexual Activity and contraceptive knowledge and use among in-school adolescents in Nigeria. International Family Planning Perspectives. 1997 (3); 23:28-33.

For Textbook

Surname, name of textbook, town & publication. Year. Edition number should be included (where appropriate) E.g. Williams DS., Textbook of Surgery, Texas: USA: Allen's publishers; 1996. Another example Nwanozie H., Akubue N., Ukoha I. et al, Engineering Technology, Owerri: Mems publishers; 2020

c. American Psychological Association (A.P.A.)

APA format is the style adopted by the nursing and midwifery council of Nigeria and many other institutions in Nigeria. The author in the text is cited thus: e.g., Osuala (1989), before or after the supposed Statement made by Osuala. It is not given number in the citation.

The reference at the end of the study goes thus:

For Textbook

If three (3) to seven (7) authors - List all the names accordingly. Surname, year of publication, name of textbook, town and publication. Eg.
Osuala Agnes, (1989) "A text book of public health nursing" Onitsha, Innosco printing press.
Adams, M., Simon, K., Jelly, B., & Kevin, A. (2001). *How to get rich faster*. Kuala Lumpur, Malaysia: MPH Publishing.

If more than 7 authors - List the first six authors, and the last author - No "&" E.g.

Okafor, F., Chioma, M., Angeli, L., Madu, A., Sampson, J. A., Thomas, S., .. Ezirim, L. (2021). *The simple way to heaven*. New York: Thomson Learning.

Note Italicizing is optional.

For Journal

Last name, initial. (Year). Title of the article. *Name of the journal*, Volume(Number), pages.ie

Surname, initials, year of publication, Title, name of magazine, volume, number and pages involved.

E.g., Osuala, A. E. (2001) Management of Childhood Diseases Nursing Outlook, volume 2. No. 5, pgs. 16-26.

Mickey, K. (2011). Becoming a nurse. *Journal of Nursing*, 13(2), 29-82.

Obika, M., & Kim, B. (2012). Thought Progression. *Journal of Psychology*, 113(3), 134-182

The volume may be ignored and number put in parenthesis and according to the 8th version of APA thus the above reference should be modified as Osuala, A.E (2001). Management of Childhood Diseases Nursing Outlook, 2 (5): 16-26.

Web Pages

(when no date of publication is given, the retrieved date is needed)
E.g. Ohiaeri, M. (Year). *Title of the webpage*. Retrieved from

<http://www.website.com>.

Nnadi, R. (2011). *The world is flat*. Retrieved from <http://www.psychology.com/worldisflat.html>.

Emmanuel, F. (n.d.). *Title of the webpage*. Retrieved DATE from <http://www.website.com>.

Iroka, M. (n.d.). The world is flat. Retrieved September 13, 2012 from <http://www.psychology.com/worldisflat.html>.

NOTE: The APA referencing style changes with every edition. Researchers are therefore advised to down load latest edition at the time of their study and apply according to update.



CHAPTER 4

RESEARCH METHODOLOGY

Data Collection

The quality of any research project/study depends on data collection. It is therefore necessary to use appropriate techniques for data collection. The technique used depends on the resources, money, and material.

Sources of Data

1. Primary Data
2. Secondary Data

- **Primary Data**

These are collected directly from the subjects being studied, e.g. their demographic data, opinions, responses, biometric data such as pulse, temperature, height, weight, and head circumference. This is more accurate, up to date and relevant. It has the disadvantages of being labour intensive and more expensive. The measurement can also be taken through a questionnaire or interview schedules, or observation guided by checklist. It is important to remind the researcher that when taking measurement on subjects, they must be accurate and reliable otherwise they will not be useful. One should guard against observation variations as

(a) Intra-Observation Variation

(b) Inter-Observation Variation

Intra-Observation Variation: This is a difference in the readings from same measurement taken by the same researcher.

Inter-Observation Variation: This is a difference in the readings of same measurement taken by the different researchers. Errors may be due to the instrument being used, e.g. weighing scale or unskilled or untrained research assistant.

Prevention of Observation Variations: Inter-observation variations can be prevented by standardizing the procedure and instrument to be used. Nothing much can be done in intra-observers variation as the disparity is from the observer..

- **Secondary Data**

These are already existing data that have been collected routinely from health and health related institutions and from the field of study for other purposes.

Sources of Secondary Data

(a) **Population Census:** This is the enumeration of the entire population residing in a defined geographical area at a particular point in time. Demographic variables obtained from the population census are sex, age, area of residence, occupation etc.

Uses of Census Figures

1. For planning health programme and distribution of resources.
2. To determine denominators of health e.g. the denominator is used for calculating the various mortality or morbidity rates.
3. Census figure can also be used for administrative and political purposes.

There are two types of population census

(i) Defacto

(ii) Dejure

• Defacto

This involves enumeration of persons that are physically present in an area. It includes visitors and temporary residents. This is the popular population census enumeration in Nigeria.

• Dejure

This type of enumeration involves persons that are normally or permanently resident in an area. This involves the count of all permanent residents who habitually live in an area. It can also be referred to as the 'resident' population. Only indigenes count. In developing countries e.g. Nigeria, census figure are unreliable because of under and over enumeration. Many a time planning and distribution of resources by African countries is based on estimated population.

(b) Mortality Data

The information on mortality is got from death certificate and birth registration records. Death certificate usually contain the name of the deceased, place of death, cause of death, ideally this should be registered with the local government commission. Accuracy of the data depends on the competence of the person who recorded the death and diagnostic facilities as well as the post mortem available at the time. In developing countries most deaths are not recorded, making information on mortality rate unreliable.

(c) Morbidity Data

These are concerned with illness and diseases in the community. They are gathered from the health institutions; hospitals, Primary Health Care facilities, laboratory and Out Patient Department Attendances, in-patient admission, discharges that are made. Other sources are the notification of diseases register. The register is usually kept on infectious diseases, industrial diseases and accidents records. Notification of disease means reporting the occurrence of certain diseases (notifiable) to appropriate quarters so that prompt and adequate action will be taken to control these diseases. There are four internationally notifiable diseases namely, cholera, yellow fever, typhoid and plague. The new emerging infectious diseases as Ebola, Lassa fever and Covid -19 has been added to the list. Routinely on monthly basis, notifiable diseases

are reported to the State for onward process to the national level. Morbidity data are further broken down into:

- (i) **Emergency notification of disease:** - These are usually the epidemic diseases.
- (ii) **Disease Register:** - Another source of morbidity data is the disease register.

This register is for certain diseases such as; cancer, tuberculosis, congenital malformation etc. The record in this register is used for early identification and follows up of these patients.

(d) **Occupational Health Records:** These are found in work places e.g., industries. Some industries prescribe medical examination for their workers, as pre or periodical medical examination and sickness or absence records are also kept.

(e) **School Health Records:** This is concerned with health of the school children. Medical examinations are conducted e.g. pre-entry, periodic and on graduation. Records of sickness, absenteeism, immunization records, etc. and treatment that are given to the children are also kept.

(f) **Insurance Company Records:** Insurance company policies contain health information of the subjects e.g. health problem that the person may have, their clinical attendances, etc. The information is used to settle one's bill in cases of accidents or other terms in the agreement.

NOTE: Data can be quantitative or qualitative in nature.

Qualitative Data: There is no notion or magnitude attached to the variable e.g., male or female. One is either a male or a female. It is a whole number; man cannot be cut into half so there is no fraction for a qualitative data.

Quantitative Data: Quantitative data on the other hand has notion and magnitude attached to the variable E.g. 7.5m, 98°F etc. Height or weight of an individual for instance can come in fraction. One's height may be 1.6m while the temperature can be 98.4°F or 37.2°C, as the case may be.

Rules to be observed during data collection

It is important for a researcher to know that he or she must:

- 1) Respect the rights of subjects.
- 2) Protect the rights and safety of subjects.
- 3) Maintain confidentiality.
- 4) Explain purpose and nature of study to subject.
- 5) Obtain free and informed consent.
- 6) Possess necessary knowledge and skill for the particular study design adopted.
- 7) Be aware of his/her limitations.
- 8) Be honest in the research use and purpose.
- 9) Have the variables in mind

VARIABLES AND ATTRIBUTES (Definition)

A variable is a quantity, which changes in value from subject to subject. It is an entity, which can take different values e.g. weight, weather, etc. are variables. It changes in a subject from time to time. Categories of variables can be quantitative or qualitative as explained earlier. A variable can also be classified as:

- a. Continuous.
- b. Discrete

Continuous variable: This is the variable that can assume either an integral or fractional values and can be measured in different units, e.g. height (m), blood pressure (mmhg), temperature in °F or °C. etc. A variable is continuous if it does not have a minimum size unit. A continuous variable is a variable that theoretically can assume any value within some specific range. Measurement of continuous variable can be approximated unlike the discrete.

Discrete Variable: Takes countable or whole number e.g. parity, white blood cell count, age on a birthday, sex or gender, etc. They are variables that are always exact because the data are composed of indivisible units.

Types or other classifications of variables

Independent Variable: A variable that affects other variables. It is the cause or the treatment applied. It may be in form of introduced programme.

Dependent Variable: This is the outcome or effect of another variable.

E.g. occurrence of cancer of the lungs is dependent on prolonged smoking. The disease cancer (effect) is the dependent variable while smoking (cause) is the independent variable.

Mediating or Extraneous Variable: This is a variable that can affect another variable and at the same time can be affected by another variable, it is another independent variable (cause) that can affect the dependent variable even though it was not originally included or intended to influence the outcome of the study. E.g. Sooth in the environment (extraneous variable) may influence or be the cause of the lung cancer but it is not the independent variable (smoking) that is being studied in relation to lung cancer in the subject.

Another aspect of a variable is the attribute. Qualitative variables have attributes

Attribute:

Attribute is a characteristic of something. It is a characteristic to which a subject belongs or does not belong, e.g. Blood group to which one belongs. Blood group is a variable and individuals can belong to Group A, B or O as the case may be. Marital status of an individual even though a variable has an attribute so also educational level of an individual etc. all have attributes.

THE METHODS OF DATA COLLECTION

There are various methods of data collection from respondents. They include:

- a) **Observation Method:** This method can be defined as a method of scientific enquiry. This involves regular and systematic observation of event and experiment. It is used quite often in the pure and physical sciences. Individuals can also be observed depending on the researcher's intent.

Advantages of Observation Method

- (i) On the spot information.
- (ii) It is reliable.
- (iii) It is less expensive.
- (iv) It is best used as a follow-up to interviewing method.

Disadvantages of Observation Method

- (i) It is suitable for only small fractions of the items to be studied,

- (ii) It is not always easy to combine observation method with random sampling.
- (iii) It gives too much scope to the influence of the individual researchers as it can be subjective.
- (iv) It is not a good method for the study of opinion and attitude.
- (v) It is an extremely time-consuming technique.

- b) **Personal Interview Method:** This is the best method of collecting statistical information. It involves an enumerator interviewing people in relation to the subject matter being studied. Interviews should be arranged well in advance and the conduct planned by the researcher.

Advantages of Personal Interview Method

- (i) Data collected can be very reliable.
- (ii) It yields a high percentage of acceptance return since enumerators can easily correct contradictory statement.
- (iii) The refusal rate is low.
- (iv) Respondents are likely to give immediate answers.
- (v) The questionnaires can be longer than the ones used in any other method of data collection.

Disadvantages of Interview Method

- (i) The cost per interview is usually high due to enumerators feeding and traveling expenses.

- (ii) Enumerators can influence answers through personal contact.
- (ii) There is possibility of enumerator not recording responses accurately.

c) Mail Questionnaires Method: This method of collecting data is one in which a questionnaire is posted to an individual to complete and returned later to the researcher. Care should be taken during its design to ensure that questions are unambiguous. A covering letter should accompany the questionnaire explaining the purpose as well as any instructions for its completion.

Advantages of Mail Questionnaires Method

- (i) A wide geographical area can be covered.
- (ii) It is a cheaper method of collecting data than personal interview method.
- (iii) Only possible method of collecting data when the population to be covered is widely spread and the time and money available is small.
- (iv) It eliminates the problems of enumerators going back to the respondents several times.
- v) It may give more accurate information on household data since individuals may take a greater care in filling the forms.

Disadvantages of Mail Questionnaires Method

- (i) Many respondents may not return the completed forms so that there is a less adequate coverage,
- (ii) The data collected may not be very representative when very few responses come from a particular area,
- (iii) There are delays in returning completed forms.
- (iv) There is no control over the return of completed form.
- v. There is the risk of group response when individual response is required.

d) Telephone Method: This is the method by which telephone calls are used to collect information from a selected sample or population of telephone subscribers.

Advantages of Telephone Method

- (i) It yields on the spot response.
- (ii) Best used in viewing habits and opinion.

Disadvantages of Telephone Method

- (i) It can lead to biased results.
- (ii) It involves only telephone subscribers.
- (iii) It is not useful for most African countries where telephone services are inefficient and only a small proportion of the population have access to telephone.
- (iv.) Data subscription and poor network may pose challenges.

e) Documentary Sources: This method requires the investigator to consult existing reports, manuals, files, journals of past and present researchers, etc.

Advantages of Documentary Sources

- (i) It is less expensive,
- (ii) Saves time.
- (iii) Data collected can supplement information obtained from other methods.

Disadvantages of Documentary Sources

- (i) Data collected may be outdated.
- (ii) Data may contain inaccurate information.

MEASURING SCALES OF DATA

Data is measured using some principal scales.

Nominal Scale: This can also be called classificatory scale. In this type of measuring scale, tags and labels are used to distinguish one measurement from the other. This scale of measurement is used for qualitative variables, using some properties (attributes) which they possess e.g. In measuring Genotype, the biological attributes are considered; AA, AS, Aa; Fertility: fertile/sterile; State of health: well/sick, Gender: male/female, Livelihood; alive/dead, all may have nominal scale applied to the data, (see preceding under analysis)

Ordinal Scale: This is also known as Ranking scale. It has the characteristics of a nominal scale with an implicit order relationship.

The classification may be relative rather than quantitative e.g. social status of a group may be stated as high, moderate or low. Condition of a patient may be ranked or ordered as unimproved, improved, or much improved. Numbers can be used to denote the relative magnitude of one to the others. It can be highest to lowest or lowest to highest.

Interval Scale: An interval scale is characterized by a numerical unit of measurement, such that the difference between any two measurements is explicitly known in terms of an interval between two points on the scale. The measurement can be ranked or ordered as well as maintaining a constant interval between any two measurements. Unlike the nominal and ordinal scale, the interval scale is quantitative. Examples temperature scale, the Celsius or centigrade ($^{\circ}\text{C}$) and the Fahrenheit ($^{\circ}\text{F}$).

Ratio Scale: This is a measurement scale with a constant interval and a true zero point. Examples of data on which this scale can be employed include heights, weights, rates, ratio etc. Many statistical procedures can be applied to either interval or ratio scales.

Likert Scale: This is a measured scale in which the degree of agreement or disagreement is noted. Strongly agree, agree, not decided, disagree, and strongly disagree; high, moderate and low are examples.

TOOLS FOR RESEARCH/DATA COLLECTION

- (a) **Mechanical:** Tape recorders, cameras, videotapes, thermometers, rulers, microscope, telescopes, etc.
- (b) **Clerical:** (Paper work) Client record e.g. bed head ticket (BHT), case studies, and interview schedules, questionnaires.

Components or Properties of Tools for data collection

It must be valid and reliable.

Validity: Whenever it is used, it must produce the same answer i.e. validity is the ability of a tool to test what it is designed to test.

Validity is the accuracy with which a test instrument measures what it is supposed to measure. The test must measure what it is expected and designed to measure. The attributes measured must be relevant to the behaviour that is to be judged.

Types/ Methods of Obtaining Validity of a tool:

(a) Concurrent Validity:

Here, one requires a criterion with which to compare the test scores, e.g. to measure validity of nursing ability of students, we require scores by the same learners on a known measure of nursing competence.

(b) Predictive Validity:

Here, criterion data are obtained months or years of the administration of the test whose validity is in question.

(c) Content Validity:

In designing a test, the teacher re-examines the test to ensure that it truly represents the course outline. The test is said to be valid depending on the extent to which it appears to be representative of the content. Content validity is determined by the relevance of the items to the objective of instruction. Content validity is tested/ established by consensus of experts in the subject area, in form of conference.

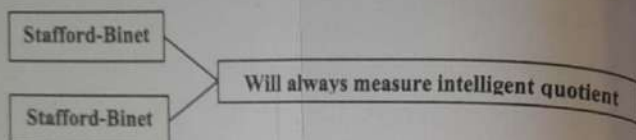
(d) **Face validity:** Refers to the appearance of a tool or test items. Judged subjectively.

(e) Construct Validity:

It refers to the extent to which a test describes a hypothesized trait. Whenever trait has not been identified, the criterion data are not readily available.

Inferences about construct validity are made from the differences in performance between the groups assumed to possess the trait and groups that do not possess the trait. For a test to be truly valid, it must be also reliable.

Reliability: The ability of a tool to consistently measure that which it is supposed to measure. Reliability deals with the trustworthiness and consistency of a measure. For intelligent test, Stanford Binet and Wechsler Bellview will always measure IQ.



Reliability refers to the degree of consistency and stability of the measures obtained from the instrument. If individuals tend to maintain the same order of merit on each of the two administrations of a test, the test is said to be reliable.

Types/Methods of estimating Reliability:

(a) **Test-Retest:** This involves administering the same test to the same learner twice and the marks obtained are correlated. The index is the "Stability co-efficient"

(b) **Parallel-test Method:**

In this method, the candidates are given two similar tests but not identical - as close together in time as possible. The sets of marks are correlated. The index obtained is called "Co-efficient of equivalence." The advantage is that it eliminates the possibility of memory defects inherent in test-retest. However in practice, two similar tests in teaching learning situation are not easy to design.

(c) **Split-Half Method:**

One test is given to learners, but divided into two halves (e.g., odd scores are thus provided and these are correlated with even scores. Index obtained is called "Internal consistency."

(d) **Reader Reliability:**

This usually applies to essay questions. Two or more examiners are used to mark the same test. This is done independently. When both or more readers come up with similar score, the test is reliable.

Factors That Affect Reliability of Instrument

The factors include inherent variation of instrument. Instruments for measurement vary and this may affect the result of findings. Fluctuation in the substance being measured could be another factor. For instance, urine that is acidic may turn to alkaline after prolonged standing. There could be observer error. The observer may make mistake in his or her readings. Another factor is inter-observer error. The observers may give two different readings for one subject or substance being observed.

Components of Validity in Epidemiological Research

Validity as has already been stated is indicative of the substance it is supposed to measure. Example, fever is a valid indicator of malaria in an area where malaria is endemic like Nigeria. The first diagnosis of a person with fever is malaria until otherwise indicated by subsequent laboratory investigations unlike in developed countries where malaria is rarely existent.

The two (2) components of validity are:

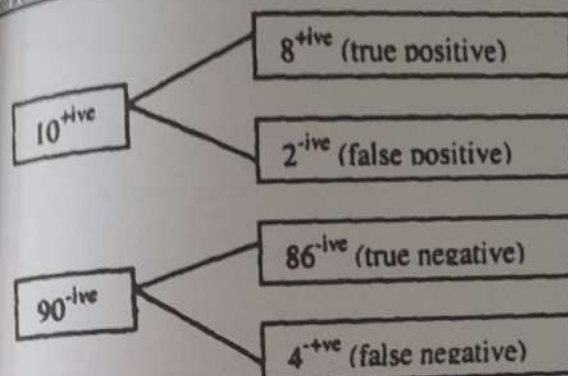
(i) Sensitivity

(ii) Specificity

Sensitivity: of a test or a particular procedure or measuring instrument is the quotient of a change in an observational measure and the corresponding change in value of the quantity of factor that is being measured. The larger the value of the quotient; the greater the sensitivity. In epidemiology, sensitivity is defined as the proportion of true positives correctly identified by a test to the sum of true positives and false negatives identified by the test. It stresses on the ability of a test to detect correctly the people that have the disease. Pregnancy test may be conducted on suspected respondents. They may all be pregnant even though the test may give a negative result to some.

$$\text{Sensitivity} = \frac{A}{A + C}$$

Where A = the number of true positives identified by the test and C the number of false negatives identified by the test. Suppose 100 patients suspected to be pregnant are sent for pregnancy test and 10 read positive while 90 read negative initially. Then a confirmatory test was conducted. Out of the 10 that read positive, two (2) now read false positive, whereas out of 90 that read negative, 4 now read false negative. It therefore shows that out of the 100 patients a total of 12 are truly positive.



Applying this formula

$$A = 8$$

$$C = 4$$

Sensitivity, which is always expressed in percentage (%) =

$$\frac{8}{8 + 4} = \frac{8}{12} * 100 = 66.7\%$$

Specificity in epidemiology on the other hand lays emphasis on the ability of a test to detect correctly those who do not have the disease or problem.

$$\frac{D}{B + D}$$

Where D = number of true negative correctly identified by the test while B is the number of false positives correctly identified by the test.

Applying the formula

$$\frac{D}{B+D} = \frac{86}{86+2} = \frac{86}{88} * 100$$

$$= 99.7\%$$

Other components of validity are **positive predictive value** and **negative predictive value**. The positive predictive value is the probability that a positive result in a test is truly positive while the negative predictive value is the probability that a negative result is truly negative.

Formula:

$$\text{+ive predictive value} = \frac{A}{A+B}$$

$$\text{-ive predictive Value} = \frac{D}{A+D}$$

QUESTIONNAIRE

Questionnaire is one of the instruments used in collecting data. Questionnaire is a set of written questions used to collect data for research based on the aims and objectives of the study. It is advisable to start formulating questionnaires ahead of time so that necessary corrections can be made.

Attributes of a Questionnaire

Characteristics of the subjects are considered: age, sex, marital status, race, ethnic group, religion, occupation, education, socio-economic status, family size, personal habit - alcohol and tobacco.

etc. All these are called demographic variables. Subjects should have similar characteristics especially in terms of location, culture etc.

There should be no geographical or time differences occurring in the given community or group (observe the above attributes and note if there is any difference in the two groups). The group must have something in common. If age affects diseases in any community, there must be age reference. Sex affects disease occurrence because certain diseases are peculiar to some sex e.g. cancer of the prostate gland is peculiar to men while ovarian cancer is peculiar to women. Disease occurrence may be more severe in certain sexes. The reason might be because of differences in the use of health facilities e.g. females make use of health facilities more than males. If the subjects do not share a common characteristics, this may produce a false result, i.e. the result of the study will become biased. Marital status has also shown that morbidity is high in people who are separated or widowed. The argument goes up and down, sideways and diagonal. The question asked is which leads to the other. One's income may be related to his occupation. This subject is treated in medical sociology. One's income places one into a particular class. This depends on the remuneration. If the topic is stated in form of hypothesis, give explanation e.g., why is it that a particular problem is found between a particular age and sex? Give reason for the relationship between age/sex.

Some environmental factors about the subjects are considered in the construction of the questionnaire. Environment can be divided into Physical, Biological and Social. In this case we start with the geographical description of the place.

(a) Physical Environment

This describes the kind of vegetation, including rivers. Physical environment refers to the non-living part of the environment - the air, soil structure, water, minerals, the temperature, humidity and rainfall, etc. The physical environment is extremely variable in the tropics; covering arid deserts, savannahs, upland jungle, cold, dry or humid plateau, marshlands, high mountain, steeps or tropical rain forest.

(b) Biological Environment

Vegetation determines the type of microorganisms found in an area. All the living things in the area - the plants, animals and microorganisms - constitute the biological environment. All these living things are interdependent on each other and they are ultimately dependent on their physical environment. Man deliberately manipulates the biological environment. He cultivates useful plants to provide food, clothing and shelter, and he raises farm animals for their meat, milk, leather, wool and other useful

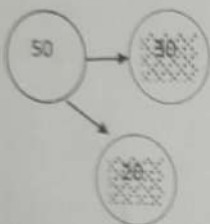
products. He hunts and kills wild animals and other predator, and destroys insects, which transmit diseases as well as vectors. This is partly because; the natural environment favours their survival but also because, in some of these areas, relatively little has been done to control these agents.

(c) Social Environment

This represents the situation of man as a member of society, his family group, his village or urban community, his culture, tradition including beliefs and attitudes, the organization of society, politics, government, laws, and the judicial system, educational system, transport and communication, social services including health services. There is much variation in the extent of technical development in the various countries in the tropics. Some of these countries are now highly developed, whilst others are still in the early stages of technical development. Some of the underdeveloped countries show certain common features - limited central organization of services, scattered population, living in small self-contained units, and inadequate control of common agents of disease. Some of these communities are still held tightly in the various circles of ignorance, poverty and disease.

Place

This can be described with a Spot Map.



This represents 50 people in the population that have the disease.

Place, answers the question of where the persons with health problems are located. The frequency of disease often varies by geographical location e.g., climate, urbanization, climatic zones, country, region, state, district, water supply, presence or absence of vectors. When considering a place, the environmental hazards in that area should be taken cognizance of. Malaria is located in tropical areas. Abakiliki because of lack of adequate clean water supply was once Guineaworm ridden.

Time

Time variation in the occurrence of a particular disease is important. Some diseases are now on the increase, e.g., coronary heart disease. Using descriptive epidemiological design you can formulate a hypothesis. Remember hypothesis may be real or apparent, marked or improved.

The importance of hypothesis is that it gives idea as to what is going on. Disease occurrence may be described in terms of hours, days, weeks, months or years. Period of time can be stated in hours, weeks, months, and years. The time can be secular or cyclic or be stated in decades. This is best described using a graph. Sexually transmitted disease may be acute on Mondays, which portrays what happens during the weekend. The researcher should find out what is common with that particular time in the year; is it rainy or dry season that favours the occurrence of the disease? An example is Onchocerciasis, which is seen during the fishing season. Incidence of Guineaworm is low between June and July. Diarrhoea and vomiting in children are more common after festivities as Easter, Christmas and Sallah celebrations. From the findings interventions are provided in terms of hypotheses.

Rules to be observed in Setting Questionnaires

A questionnaire should be kept as brief as possible and only relevant questions should be included. These questions should be clear and unambiguous. Technical jargons should be avoided as well as leading questions. The question should be arranged in sections and follow each other logically. Sensitive questions may be self or interviewer administered. The self-administered method is cheaper because it takes less time to administer. It can even be sent and collected by mail. It is also useful when the study population is large

and scattered over a large population area. It has the disadvantage of low response. The interviewer administration of questionnaire is more expensive but the response rate is higher.

Package of Questionnaire: Consists of 6 items

- i. Covering letter.
- ii. Instruction to respondent.
- iii. The questions.
- iv. Response, categories and precoding.
- v. Demographic information.
- vi. Inducement to respondent.

i) **Covering letter:** The covering letter tells the respondent who you are, the purpose of the paper and what you hope to get out of it. The inducement will also be in this coverage.

ii) **Instruction to respondent:** The letter of instruction will tell the respondent the details of completing the questionnaires.

iii) **The Questions:** The questions in the questionnaire must be examined for appropriateness of content, wording and order. The content deals with the specified topic of your study. It must address the objective of the study.

Wording: In wording a question, four characteristics must be considered:

1. Clarity
2. Brevity (shortness)

3. Simplicity

4. Applicability

Avoid questions that are double barreled (double negative questions, jargon filled questions). Do not use complex or biased question e.g., do you think good mothers should breastfeed? Avoid embarrassing questions. The question must be orderly i.e. move from simple to complex regardless of the method one is using; open ended or close- ended, the questions should follow a logical order with no abrupt transition.

iv) Response Categories and Precoding

When using a structured question, researcher should have already determined on the category, which he wants. Each response category must be mutually exclusive and exhaustive. Subjects must be able to choose one and only one alternative. The alternative response must be appropriate for each other. For the purpose of analysis, for variables that are qualitative, one can arbitrarily assign values in numerical form to the attributes e.g. weather is a variable. It may be cold or hot. These are attributes of weather; (1) can represent cold while (2) will be for hot. Age is a variable, being young or old is an attribute, young may assume (1) while old assumes numerical code of (2). Pre-coding can be done. It means that you assign a numerical code to the questionnaire responses especially those dealing with nominal scale e.g., sex; you can say mark (1), if male then (2) for female. Pre-coding is essential for easy analysis. There are responses to fixed alternative questions.

Three examples of response to fixed alternative question

1. Likert scale
2. Rank order (*ordinal scale*)
3. Checklist.

Likert Scale: The response is graded. One wants to know the degree or level of the subject's conviction. Likert is a measuring scale for summated rating devised by Rensis Likert. e.g. strongly agree, agree, undecided, disagree and strongly disagree.

Rank order: Researcher want subjects to rank these things in order of importance or order of less importance e.g., first being less important and last one, of most importance, as the case may be (very severe, moderately severe, slightly severe).

Checklist: Subjects are given a list of items and are expected to tick the one that applies most to the category. The subject just have to match whichever he feel is appropriate.

v) Demographical information

This deals with whom, where, and when. The "who" is concerned with person, his/her age, class, sex, religion, educational background; are stated. The "where" concerns the place; is it in the rural or urban, is the respondent from the south or west? The "when" is related to time, what time of the year, week, or month, or season does the incident occur? Remember that demographic data is not the subject of your study.

vi) Inducement to respondents

Researcher has to appeal to the respondents for their good nature e.g. tell them that their responses would make them to be remembered as the people that contributed to the success of the study even though anonymity is maintained. The size or appearance of the questionnaire will also affect the rate of inducement. People want to pick up what is colorful and not voluminous. It must be well typed. Many a time, anonymity is also an inducement.

Types of Questionnaires

Questionnaires can come in various types.

- (a) Structured questionnaire (i.e. close ended)
- (b) Unstructured questionnaire (i.e. open ended)
- (c) Semi-structured questionnaire (combination of a & b)

Structured / Standardized / Fixed Alternative

This resembles the multiple choice question (MCQ) examination, in that the respondent is limited to a list of options. It is easier and quicker to analyze. However, the answers are restricted to those available in the options.

Unstructured/Open ended

In this case the respondent provides the answers in his own words. It has the advantage of giving more detailed and in-depth information. This is why it is used during the pilot study and the

answers are now used to make the structured one. It is also used in qualitative studies where opinion and beliefs of respondents are sought for,

Semi-structured: This is mixture of both structured and unstructured.

NOTE: In filling questionnaire, the respondent must be assured confidentiality. In most cases, names, detailed addresses are not used except where further information may be needed from the respondents.

SAMPLING

Data are rarely collected from the whole population. Many times this is not feasible. Majority of studies are done with a proportion of the population, thus a process known as sampling is involved. Sample means a fraction of a target population. Whatever method is used in data collection, only a fraction of the subjects or items is used for most studies.

Major reason for using a Sample

1. **Sample is cheaper:** This is because of saving in cost of transport, accommodation, etc., involved if the whole population is to be reached, assembled or interviewed.

2. **Sample is quicker:** Only selected proportion of the study population are involved. In other words, the use of sample saves time.
3. **The only way out:** Sampling is often the only feasible method to find information about the population e.g., a research about the standard of living of nurses in Nigeria or estimating haemoglobin of a subject.
4. **Better result:** Yields better result.

Essential Steps in the Discharge of a Sample Survey

1. Define your objectives.
2. Decide how the information is to be collected. Before a sample is taken from a population, it is essential to ascertain that the field covered by the data is wide enough.
3. Prepare the questionnaire.
4. Select the sample.
5. Undertake a pilot survey.
6. Enlighten and brief the investigator/research assistant (if any is being used).
7. Organize the data collected.
8. Analyze and interpret the data.
9. Present the data collected in useful forms.
10. Write the reports.

TYPES OF SAMPLE

There are various sampling techniques in selecting a sample from a population: probability and non-probability sampling. In probability sampling, every unit of the study (large) population has an equal chance or probability of being selected. While in non-probability sampling, every unit does not have an equal chance or probability of being selected. Sample chosen depends on the wishes of the researcher. This depends on her objective, design and minimum sample size for the study.

- (i) **Random Sampling:** (sometimes known as probability sampling). In this type of sampling each item in the population has an equal probability of being selected. Random does not mean haphazard. Random selection is determined objectively, usually by means of the table of random numbers or by computer generated random numbers.
- (ii) **Systematic Sampling:** (Quasi-random sampling)- This sampling method requires every n th item in the list of the population to be selected by some random method. E.g., if the population is 200,000 and 10,000 is the sample required, under the systematic sampling 200,000 is divided by 10,000 giving 20. This means that in every 20, a unit is selected. The crucial question is determining the starting point. If for

example, 11 turns up, the eleventh item in the list is the first, the thirty first and fifty first are second and third samples selected respectively.

Stratified Sampling (Representative sampling): under this method, the parent population is divided into strata, group or classes, and a random sample is taken from each strata.

Purposive Sampling or Convenience: The investigator may choose an appropriate sample by conscious selection. Under this method, the probability of personal bias occurring is very high and the probability of the error cannot be calculated.

Multi-Stage Sampling: The sample is prepared by stages. The universe is divided into a number of large sampling units, each of which in turn is divided into smaller units, and so on. E.g. if a researcher is conducting a study for a targeted population in Nigeria, he may have his first stage of selection as randomly choosing a geographical zone. Next stage may be selecting a State in that zone, then a Local Government in the State before villages or wards as the case may be. Thus it is known as multi stage.

Quota Sampling: In this method, the investigator does not have list of the names and addresses of the respondents, but is asked to interview individual who have the required features.

(vii) **Sequential Sampling:** This method requires us to conduct the experiment one at a time. Whether the research is to be continued depends on the results obtained in the previous experiment. In other words, we say sequential sampling is a technique by which a sample is built up one item at a time. This technique involves the processes of accepting, rejecting or continuous inspection until the batch is either accepted or rejected. This process continues until required number is got.

(viii) **Cluster Sampling:** This involves a group with same characteristics. It is useful if one wants to select the study population from a wide geographical area. It is a sub unit of the study population found in the same geographical area e.g., individuals in the village, work place, children in school etc. A map is usually designed for use in this type of sampling. One can make a list of all the villages in a Local Government Area and then randomly select the required number of villages for the study.

(ix) **Snow Ball Sampling:** This is done when the variable under investigation is difficult to locate or has some social stigmatization e.g., Teenage pregnancy, Tuberculosis, Leprosy etc. The 1st individual is selected and he/she then helps in identifying others. This process is repeated until the required number for the study is achieved.

ESTIMATION OF SAMPLE SIZE

Every research study demands adequate sample- size that could represent the entire population. If too small, the result will not be valid and one cannot use it to generalize. The bigger the size is, the better the result. If the sample size is too large, it becomes cumbersome, cost intensive, time wasting, and resources/money are wasted. Statistical formula therefore is used to calculate, desirable sample sizes for various types of studies. The calculation can be done manually or computer may be used. One can use a documented prevalence of a particular disease or incident in the calculation or assume a certain percentage of those concerned with the issue under study. For instance 40% of a population below 100,000 may be considered appropriate as appropriate sample size for a study. Above this figure should be computed using a scientific formula as such number is quite significant. Some scientific and standard formulae adopted in sample size computation are stated below.

Examples of formulae are:

$$(i) \quad N = \frac{Z^2 P(1-P)}{d^2}$$

Where:

N = Sample size

Z = Constant of 1.96²

P = Prevalence ie., Percentage of those affected with a particular disease
d = Standard deviation of 0.5

Assuming we are to calculate the sample size in a study concerning a disease with a 20% prevalence

Using the above formula

$$Z = 1.96^2$$

$$P = 20\% = 0.2$$

$$d = 0.05 = 0.05^2 = 0.0025$$

$$1 - P = 1 - 0.2 = 0.8$$

Substituting into the formula:

$$N = \frac{1.96^2 \cdot 0.2(1-0.2)}{0.0025}$$

$$= \frac{3.84 \cdot 0.2(0.8)}{0.0025}$$

$$= \frac{3.84 \cdot 0.16}{0.0025} = 245.7$$

Minimum Sample Size = 246 (Yamane, 1973).

(ii) Another formula is the one using standard error. This is used for a smaller sample size (Maxwell, 1998).

$$N = \frac{P(100 - P)}{SE^2}$$

Where:

P = Prevalence

N = Sample size

SE = Standard error = 5

If the prevalence is 20% as above, substituting into formula

$$N = 20 \frac{(100 - 20)}{5^2}$$

$$= 20 \frac{(80)}{25} = \frac{1600}{25} = 64$$

Minimum sample size using this formula is 64.

Formula for two proportions (Maxwell, 1998).

$$n = \frac{[Z_{\alpha} \sqrt{2 p_c (1-p_c)} - Z_{\beta} \sqrt{p_t(1-p_t) + p_c(1-p_c)}]^2}{(p_t - p_c)^2}$$

Where n = minimum sample size required.

p_c = proportion in the control group.

p_t = proportion in the intervention group

α = level of significance

β = type of error

Z_{α} = standard normal deviate

Z_{β} = Z score corresponding to the alternate hypothesis.

$\alpha = 5\% = Z_{\alpha} = 1.96$ $\beta = 1.0\%$, $Z_{\beta} = 0.84$

Standard normal deviate corresponding to a power of 80% = 0.84 will apply.

$$P_t - P_c = 10\% \quad (0.1)$$

iii) Krejcie Morgan sample size formula (1970)

$$S = X^2 NP(1-P) / d^2(N-1) + X^2 P(1-P)$$

Where:

S = Required sample size

X^2 = The table value of Chi-Square for 1 degree of Freedom at the desired confidence level (3.841)

N = The population size

P = The population proportion (assumed to be .50 since this would provide the maximum sample size)

D = The degree of accuracy expressed as a proportion (.05)

Advantages of Sampling

1. Economical in terms of money, personnel and resources.
2. Follow up of non-responses is easier.
3. It allows for the use of high quality interviewers.
4. Less time consuming thus, result is out within a minimum period.
5. Sample may be the only possible way of the measurement e.g., blood sample; if all the blood is drained, the subject dies.

Disadvantages

1. **Sampling error:** There could be discrepancy between sample result and what the whole population result would have been.

2. **Discrimination:** A feeling of discrimination may be aroused in the target population.

SAMPLING ERRORS

There are usually two types:

1. Sampling Error
2. Non-Sampling Error

A **Sampling error** is an error inherent in sample studies because of the fact that one has only selected or observed only what he intended to select or observe. Whether we use a known method of selecting our sample or not, there must be a sample error. The amount of error will depend on the sample size, the greater the sample size, the lesser the error. In probability sampling (random) we can quantify the amount of sampling error using the formula:

$$SE(\bar{X}) = S / \sqrt{n}$$

Where: SE = Sample error

$\bar{X}$ = Arithmetic mean

S = Standard deviation

n = Sample size

Note: The implication is that if the standard deviation increases, the sample error will also increase but when the sample size increases, the sample error will decrease or disappear as the case may be.

Non Sampling Errors

Also known as bias error. It increases as the sample size increases. Unfortunately, we cannot quantify the amount of non-sampling error as we did for the sampling error, so we try to minimize the error. Non sampling error is committed as a deliberate selection of sample units e.g. if one cannot reach the actual people on time, in accessibility or otherwise, hand picking is done for convenience. Non-sampling error should be discouraged to avoid bias.

CAUSES OF BIAS

1. Replacement of units already selected.
2. Failure to recover the whole or greater percentage of the questionnaire given to the sample population.
3. Unskilled personnel e.g. interviewers, enumerators, etc.
4. Lack of response by the study population especially if advocacy approach was not adopted.
5. Calculation error, especially during analysis whether it is done manually or even by computer, error may come through putting down 85 when you have 58 in mind or feeding the computer with 34 when one means 43.

PILOT STUDY

There is what is called pilot study. Pilot study is done before the main study. The questionnaires are given to a similar community

to be studied. It is a miniature of the main study as it involves the whole process. This is useful to test the instruments, options to be used in questionnaires. The questions are unstructured and open ended. The ideas or options provided by the respondents are used in the construction of the structured questionnaire for the main study.

LIMITATIONS OF RESEARCH

As every endeavor in life has its flaws, so also is the researcher faced with obstacles during the study. These obstacles are referred to as limitations. Some researchers may experience limitations in their study as a result of:

1. Limited Ability

The researcher may find himself lacking in knowledge, skill or both, needed for the conducting of the study to a logical end.

2. The moral or Ethical Consideration

The moral or Ethical Consideration related to certain types of study may be a hindrance. In testing the effectiveness of a poison, one may not be justified in using human beings. If a man is to study some group of Moslem women, he will be limited to only those who are not under 'Purdah'.

3. Human complexity

It is easy to study the physical and social aspect of one's subjects but when the study is related to the psychological aspect of

man, there could be limitations. A woman may be smiling and giving a response that all is well, shortly after a serious fight and thorough beating by her husband behind closed doors.

4. Unavailability of Literature

Many a time, the required literature where knowledge of previous works done by researchers may not be available especially if one finds himself in a business community where the main language is "money". The government and the people are usually more interested in "quick cash", "share the money" rather than "publish or perish". Researches are rarely funded in such societies and publications are of little or no importance amongst such populace, hence paucity of literature.

5. Lack of Materials

The necessary tools needed for the research may not be readily available. The ones available may not be functioning well and as such provide false results. If a laboratory scientist needs dye for staining and is provided with gentian violet for the test. Outcome/result would not be same. A nurse would also find herself in similar situation if she is to check a child's blood pressure and all that is available is the adult sphygmomanometer with adult cuff

6. Time

Agencies and institutions usually prescribe the time for the duration of the study. One therefore cannot embark on a study that will take him two years to complete when he is supposed to conclude his program in the institution within six months.

7. Finance

The amount of money available for the study determines the type, duration and quality of the study. More money will be needed for an elaborate and extensive study.

CHAPTER 5

DATA ANALYSIS

After information or data have been collected, they are sorted out and analyzed using the computer or manually. Data analysis is the process by which the researcher summarizes and describes the data collected. When possible, he makes inferences from the larger population from which the sample came from. Analysis of data is done using either descriptive or inferential statistics. At times both may have to be employed. Descriptive statistics is used to summarize and describe the data while inferential statistics is used to estimate the probability that what the researcher found from the study sample will be generalized to the target population. Inferential statistics refers to testing the hypotheses. Researchers use descriptive statistics more often.

The form of descriptive or inferential statistics used, will depend on the aim or funding of the study, coupled with the capability of the researcher. Analysis can be done using measures of central tendency as mean, media, and mode. Measures of variation, which include range, variance, standard deviation, and coefficient of variation may be adopted for analysis depending on the researcher's

choice. Percentage, ratio, proportion and even the quartile can, also be used. All these are termed, descriptive statistics.

Pre-coding of the items on the questionnaire may be necessary before analysis, especially if computer processing is employed. Coding is used to identify the response category before the questionnaire is sent out for analysis. Various type of coding can be employed such as:

1. **Numeric coding:** e.g. Married (1) Single (2) Widowed (3), e.tc
2. **Alphabetic coding:** e.g., Married (a) Single (b) Widowed (c) e.tc
3. **Alphanumeric:** Combination of alphabets with numerical e.g., Married (A_1), Single (A_2), and Widowed (A_3)

When statistical analysis is done with computer, some statistical software/packages as EPI-Info, Statistical Package of Social Sciences (SPSS) of different versions may be used. Most recent version of SPSS is always recommended.

DESCRIPTIVE ANALYSIS

The purpose of statistics is either to describe facts as they are contained in a set of data or to generalize and go beyond the information, which are supplied. The type of descriptive we use in a given problem may depend on whether we want to generalize on the basis of the sample information or we want to describe the population. Under this part, two basic kinds of description known as measures of central tendency and variation will be discussed.

A. MEASURES OF CENTRAL TENDENCY

Measures of location are often referred to as measures of central tendency, which tells us the location of central or middle of a set of data. The measures of location we shall be concerned with are:

- (i) Arithmetic mean (if no extreme value)
- (ii) Median (if extreme values are low and high)
- (iii) Mode (if value that occurs most is needed)
- (iv) Geometric mean (if extreme values are medium)
- v. Harmonic mean (if extreme value is high)

(i) MEAN

a) Arithmetic Mean (Ungrouped Data)

The arithmetic mean can be defined as the sum of all the items in a set of data divided by the number of items in the set of data i.e., summation of the items over the frequency. Given a set of numbers, X_1, X_2, X_3 the arithmetic mean denoted by $\bar{X}$ is:

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} = \frac{\sum X_i}{n}$$

Example

Find the arithmetic mean of the following: 25, 10, 15, 16 and 24.

Solution: Let X_i represent the total numbers, hence the arithmetic mean:

$$\bar{X} = \frac{25+10+15+16+24}{5} = 18$$

b) Arithmetic Mean (Ungrouped Data) using - Assumed Mean

Sometimes, the arithmetic mean for ungrouped data can be calculated using an assumed mean. This is calculated by the formula:

$$\bar{X} = A + \frac{\sum di}{n}$$

Where A = Assumed Mean, $di = X - A$ = deviation from A .

Example

Given the following marks of five students in an examination 25, 10, 15, 16 and 24. Find the Arithmetic Mean using the method of an assumed arbitrary mean.

Solution

Step 1: Write down the formula $= \bar{X} = A + \frac{\sum di}{n}$

Step 2: Assume the mean. This is best done by choosing a number between 15 and 24. Let us assume 16.

Step 3: Create the table

X	$X - A = d_i$
25	$25 - 16 = 9$
10	$10 - 16 = -6$
15	$15 - 16 = -1$
16	$16 - 16 = 0$
24	$24 - 16 = 8$

Step 4: substitute into the formula

$$\bar{X} = 16 + \frac{10}{5} = 16 + 2 = 18$$

c) Arithmetic Mean (Grouped Data)

In calculating the Arithmetic mean of the grouped data, we often use the class mark or mid-point of the class with the corresponding frequencies. If we have a frequency distribution with class intervals $x_1, x_2, x_3, \dots, x_n$ as mid points having the corresponding frequency of $f_1, f_2, f_3, \dots, f_n$. The Arithmetic Mean is

$$\bar{X} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{\sum fx}{\sum f}$$

Example: Given the following frequency distribution of the ages of 20 students in Owerre-Nkwoji Girls Secondary School.

Age	Frequency
10 - 12	6
13 - 15	8
16 - 18	4
19 - 21	2

Find the average age of the students.

Solution

Step 1: Create the table of the distribution.

Age	Frequency (f)	Mid-point (x_1)	fx_1
10 - 12	6	11	66
13 - 15	8	14	112
16 - 18	4	17	68
19 - 21	2	20	40
	$\sum f = 20$		$\sum fx_1 = 286$

Step 2: Substitute into the formula:

$$\bar{X} = \frac{286}{20} = 14.3$$

NOTE: Mid points are calculated by adding the two class limits and dividing by 2. Example: first class = $10 + 12 = 22 / 2 = 11$,

$fx_1 = f$ multiplied by x_1

d) **Arithmetic Mean (Grouped Data) Using Assumed Mean**
The Arithmetic Mean for group data can be calculated by using alternative method of assumed arbitrary mean.

$$\bar{X} = A + \frac{\sum f_i d_i}{\sum f_i}$$

Where:

A = Assumed Mean

$d_i = X_i - A$ = deviation from Arbitrary Mean

X_i = Mid points.

Example: Given the following data on age distribution of 20 students in Isunjaba Comprehensive School.

Age	10 - 12	13 - 15	16 - 18	19 - 21	Total
No of students	7	8	2	3	20

Find the average age by Assumed Mean method.

Solution

Step 1: Write down the formula:

$$\bar{X} = A + \frac{\sum f_i d_i}{\sum f_i} \quad \text{Let } A = 15$$

Step 2: Create the table:

Age	Frequency (f _i)	Mid-point (x _i)	d _i = x _i - A	f _i d _i
10 - 12	7	11	11 - 15 = -4	-28
13 - 15	8	14	14 - 15 = -1	-8
16 - 18	2	17	17 - 15 = 2	4
19 - 21	3	20	20 - 15 = 5	15
	$\sum f = 20$			$\sum f_i d_i = -17$

Step 3: Substitute into the formula:

$$\bar{X} = A + \frac{\sum f_i d_i}{\sum f_i} = \frac{15 - 17}{20} = 15 - 0.85 = 14.15$$

e) **Arithmetic Mean using the weighted Mean method**

There are various problems in which we cannot compute average quantities without paying some attention to their relative

importance in the overall situation we are trying to describe. For instance, there are two pharmacy shops in Port Harcourt selling multivite tablet at ₦80 per tin and ₦95 respectively; we cannot know the overall price paid for multivite tablet in Port Harcourt unless we know the number of tins sold by each pharmacist. If majority of consumers buy their multivite from the first pharmacist, the average price they pay per tin will be closer to ₦80 and if of them buy from the second trader, the average price will be to ₦95. Therefore, we cannot calculate a meaningful average unless we know the relative weight (relative importance) carried by the fiber we want to average.

For example, to average a set of numbers with relative importance express numerically by means of some numbers:

$W_1, W_2, W_3 \dots W_n$ which is their weights.

$$\text{The weighted mean } \bar{X} = \frac{\sum f_n}{\sum W}$$

Example 1: Assuming the first pharmacist sold 150 tins while the 2nd sold ₦120 tins, applying formula:

$$\frac{80(150) + 95(120)}{80 + 95} = \frac{12000 + 4400}{175} = \frac{13400}{175} = \text{₦76.5 per tin of multivite}$$

Example 2: A chemist bought 16 liters of detol at ₦6 per liter, 22 liters at ₦7 per liter and 25 liters at ₦9 per liter of detol. What is the average price he paid per gallon of detol?

Solution

Step 1: Determine the weights (w) and price (x):

Weights (w): $W_1 = 16$; $W_2 = 22$; $W_3 = 25$

Price (x) = $X_1 = 6$; $X_2 = 7$; $X_3 = 9$

Step 2: Write down the formula and substitute into the formula: ^

$$\bar{X} = \frac{\sum Xw}{\sum w} = \frac{6(16) + 7(22) + 9(25)}{16 + 22 + 25} = \frac{96 + 154 + 225}{63}$$

= ₦7.54 per liter

Features (Properties) of Arithmetic Mean

1. It always exists and can be calculated for any form of numerical data. It is always unique in the sense that there is only one arithmetic mean for any set of data.
3. Most people easily understand it.
4. It is arrived at by using all items in the data.
5. Arithmetic mean for frequency distribution with open classes cannot be determined except some assumptions are made.
6. It is the most useful measure of central tendency and is amenable to further mathematical analysis.

7. The sum of the deviations of the random variables from the arithmetic mean is zero.

Advantages of Arithmetic Mean

1. It is a member of the mathematical system, which is amenable to more advanced statistical analysis.
2. It includes the values of all items in the distribution.
3. It is always unique in the sense that there is only one arithmetic mean for any set of data.
4. Most people easily understand it.
5. It is the best known of the averages.

Disadvantages of Arithmetic Mean

1. The mean may not coincide with the actual value calculated.
2. Mean may be unrepresentative because of the effect of extreme values.

Example: Average age of three people: 30, 35 and 65 = 43.3. The age of third person 65 makes the mean unrepresentative.

3. It may result in an impossible figure where the data are discrete e.g. 1.83 women or 4.5 cheques.
4. It is sensitive to extreme values.

ii) THE MEDIAN

Median is a middle observation in a set of data that has been arranged in a particular order of magnitude.

a) The Median (Ungrouped Data)

The median for an ungrouped data is the value of the middle item or the arithmetic mean of the values of the two middle items arranged in an increasing order of magnitude. If we have an odd number of items, there must be middle item whose value by definition gives us the median.

Example 1: Given a set of numbers: 25, 7, 30, 42, 8, 15 and 2, find the median.

Solution: The median is obtained by re-arranging them: 2, 7, 8, 15, 25, 30, and 42. The median is the middle item, which is 15.

Example 2: Find the median of the following set of numbers: 25, 7, 30, 42, 8, 15, 2 and 50.

Solution: The median is obtained by re-arranging the numbers: 2, 7, 8, 15, 25, 30, 42 and 50. Then select the middle items. The median is:

$$\frac{15 + 25}{2} = 20$$

In general, the median for the set of odd numbers of observation of item is

$$\frac{(N+1)}{2}$$

Provided that these numbers are arranged in order of magnitude

Example: Find the median of 11, 4, 16, 9, 2, 13, and 7.

Solution

Step 1: Re-arrange the data = 2, 4, 7, 9, 11, 13, and 16.

Step 2: $N = 7$ = number of items.

$$\text{Median} = \frac{N+1}{2} = \frac{7+1}{2} = 4^{\text{th}} \text{ item}$$

The 4th item is 9, which is the median.

b) Median (Grouped Data)

The median of a grouped frequency distribution is defined as that value or potential value in the distribution of items above and below which are half of the frequency.

$$\text{Me} = L_1 + \frac{\left[\frac{\sum f}{2} - \sum f_i \right]}{f_{me}}$$

Where:

Me = Median value.

L_1 = True lower class limit of the median class.

$\sum f / 2$ = Half of total observations,

f_{me} = frequency corresponding to median class.

$\sum f_i$ = Total or cumulative frequency above media class,

i = class interval of median class.

Example: If the age distribution of women of child bearing age in Amudim in Amugbara is as follows:

Age	Frequency (f)
15 - 19	5
20 - 24	10
25 - 29	15
30 - 34	8
35 - 39	3
40 - 44	2
Total	43

Calculate the median age of the women.

Solution

Step 1: Re-write the table paying attention to the true lower and upper limits and creating a column for cumulative frequency.

Age (yrs)	Frequency (f)	Cumulative frequency (cuf)
14.5 - 19.5	5	5
19.5 - 24.5	10	15
24.5 - 29.5	15	30
29.5 - 34.5	8	38
34.5 - 39.5	3	41
39.5 - 44.5	2	43
Total	43	

NOTE: To determine the median class, it is simply the class with the highest frequency. In the above table, it can be seen that the class with 24.5 - 29.5 has the highest frequency of 15. This is the median class. Alternatively, it can be determined by the formula

$$\frac{\sum f}{2} = \frac{43}{2} = 21.5$$

21.5 lies between 15 and 30 in the cumulative frequency column. Thus, 30 with 15 as frequency and 24.5 - 29.5 are the median class.

$(\sum f)_i$ = sum of all classes above or before the median class = 5 + 10 = 15

Time = frequency of median class = 15

i = class interval = 29.5 - 24.5 = 5

U = lower class limit of median class = 24.5

Step 2: Substitute into formula:

$$\begin{aligned}
 &= 24.5 + \frac{(21.5 - 15) 5}{15} \\
 &= 24.5 + \frac{6.5}{15} \times 5 \\
 &= 24.5 + 2.17 = 26.7
 \end{aligned}$$

Features (Characteristics) of the Median

1. It is not sensitive to extreme values.
2. It always exists and is unique.

3. The median of grouped data can be found even if the distribution has open classes at both ends.
4. It can be used to find the middle item of numbers or quantities which cannot be described by numbers.

Advantages of the Median

1. It always exists and is unique.
2. It is not sensitive to extreme values. This means that median is the most useful measure of central tendency when the arithmetic mean is unacceptable.
3. The median of grouped data can be found even if the distribution has open classes at both ends.
4. It can be used to find the middle item of number or quantities which cannot be described by numbers.
5. It may be found with a minimum of mathematical calculations.

Disadvantages of the Median

1. It cannot be used for subsequent calculations as is done with the mean.
2. We must first arrange the data according to size or array of scores and this is tedious.
3. If the distribution is irregular, the calculation of the median may be indefinite.
4. Median for grouped data may not be located exactly.

5. We cannot find the overall median without going back to the raw data.

For example, the median of 6, 10, 18, is 10, the median of 11, 35, and 40 is 35, and knowing only these two medians, we would never be able to obtain the result that the overall median of the six numbers is 14.5.

(iii) MODE

a) Mode (Grouped Data)

This is the most easily determined of all measures of location. For ungrouped data, the mode is the number of item that occurs with the greatest frequency. The mode may not exist if no item repeats itself and may not be unique if more than one item repeats itself.

Example 1: Find the mode of the following set of numbers: 2, 2, 3, 4, 6, 6, 6, 9, and 9.

Solution: The mode is the number that has highest frequency or repeats itself more often. Hence the mode is 6.

Example 2: Find the mode of the scores of eight students in an examination 40, 50, 52, 61, 68, 70, 72, and 74.

Solution: None of the marks repeats itself i.e., since all the numbers have frequency of one; the scores have no modal mark.

Example 3: This data represent profits of Fenice Pharmaceuticals, Nkpor for some given number of years in thousands of naira: 420, 530, 530, 640, 640, 640, 750, 750, 870, 870, 870, 900. Find the modal profit of the company.

Solution: In this problem, there are two numbers 640 and 870 that repeated themselves three times; hence the modal profits are 640 and 870. Hence there are two modes, the data is said to be BIMODAL.

b) Mode (Grouped Data)

The mode for a grouped frequency distribution is obtained by the formula.

$$\text{Mode} = L_1 + \frac{1(\Delta_1) C}{\Delta_1 + \Delta_2}$$

Where:

L_1 = True lower limit of the modal class. The modal class like the median class is the class with the highest frequency.

Δ_1 = Excess of modal frequency over the frequency of the class before it. i.e., difference between the frequency of modal class and the frequency above it

Δ_2 = The excess of modal frequency over the frequency of the class after it i.e.; difference between the frequency of modal class and the frequency below it.

C = Class interval of modal class.

Example: Given the following frequency distribution on the weights of Chemicals from Fenice Pharmaceuticals, Nkpor.

Weight (kg)	Frequency (f)
118 - 126	5
127 - 135	8
136 - 144	3
145 - 153	16
154 - 162	2
163 - 171	9
172 - 180	7
Total	50

Calculate the mode.

Solution

Step 1: Write down the formula = $L_1 + \frac{[\Delta_1] C}{\Delta_1 + \Delta_2}$

Step 2: Rewrite the table using the true upper/lower class limits.

Weight	Frequency (f)
117.5 - 126.5	5
126.5 - 135.5	8
135.5 - 144.5	3
144.5 - 153.5	16 Modal class
153.5 - 162.5	2
162.5 - 171.5	9
171.5 - 180.5	7
	50

NOTE: $\Delta_1 = 16 - 3 = 13$ and $\Delta_2 = 16 - 2 = 14$

$$\text{Mode} = 144.5 + \frac{(13)9}{13 + 4} = 144.5 + \frac{117}{27} = 144.5 + 4.3 = 148.8$$

Characteristics (Features) of the Mode

1. It is not a commonly used measure of central tendency.
2. It may or may not be unique because there can be two or more modes in any distribution.
3. Its value is not affected by open class intervals.

Advantages of the Mode

1. Its value is not affected by open class intervals.

2. It may be unique provided that there is only one mode in the distribution.
3. The mode is the best average to use if the most typical value is needed.
4. It can be estimated from incomplete data.
5. It is not necessary to know the values of all the items in the distribution in order to calculate the mode.

Disadvantages of the Mode

1. It is not a commonly used measure of central tendency.
2. Where there is no marked clustering of values around a single point, the mode is meaningless.
3. It may not be unique because we can have more than one mode in any distribution.
4. It cannot be used for further mathematical analysis.

(iv) GEOMETRIC MEAN (G)

The Geometric Mean (G): for a given data $X_1, X_2, \dots, X_n$ is the n th root of the product of the numbers.

$$G = \sqrt[n]{(X_1)(X_2) \dots (X_n)}$$

Example: Find the Geometric mean of numbers 6, 9, 10.

$$\text{Solutions : } G = \sqrt[3]{6 \times 9 \times 10} = \sqrt[3]{540} = 8.14$$

Reasons why Geometric Mean is preferred to other measures of Central Tendency

1. Geometric mean is used mainly in problems in which we are interested in averaging ratios particularly index numbers.
2. "G" is used mainly to average positive numbers since its value could otherwise be negative or imaginary.
3. Like the Arithmetic mean, it takes into account, each individual item.
4. It is always uniquely defined.
5. It reduces to some extent the effect of a very large or very small number and this is one of the reasons why it is sometimes preferred over Arithmetic mean.

Advantages of Geometric Mean

1. It uses all the data in the group.
2. It places less weight to large items as in the case of the Arithmetic mean.
3. It is exact, provided that all quantities are greater than zero.
4. It is useful in problems in which we are interested in averaging ratios particularly index numbers.

B. MEASURES OF VARIATION

Variation is the dispersion or scatter of a set of values. A measure of the degree of dispersion of data is needed for two basic purposes.

- (a) To assess the reliability of the average of the data.
 - (b) To serve as a basis for control of the variability.
- For example, assessment of the degree of quality variation is an essential part of quality control procedures in factories.

The measures of variation under discussion here are as follows:

- i. Range
- ii. Mean Deviation
- iii. Variance
- iv. Standard deviation
- v. Coefficient of Variation
- vi. Quartile

(I) THE RANGE

This is the degree to which numerical data spread about an average value. The range is the simplest and most straightforward measure of dispersion. The range is defined as the difference between the largest and smallest numbers in the set of data.

a) The Range (Ungrouped Data)

Example: Find the range of the following set of numbers; 22, 22, 41, 26, 15, 15, 12, and 11.

Solution

Step 1: Re-write the numbers in ascending order: 11, 12, 15, 15, 22, 22, 26, and 41.

Step 2: Take the first and last number and find the difference.

$$\text{Range} = 41 - 11 = 30.$$

b) The Range (Grouped Data)

To find the range of a grouped data subtract the class mark of the lowest class from class mark of the highest class.

Example: Consider the weights of 50 members of Isu Confederated Union, Onitsha branch.

Weights (kg)	Number of Members
60 - 62	5
63 - 65	10
66 - 68	13
69 - 71	7
72 - 74	15
Total	50

Find the Range

$$\text{Solution: Class mark of highest class} = \frac{74 + 72}{2} = 73$$

$$\text{Class mark of lowest class} = \frac{60 + 62}{2} = 61$$

$$\text{Range} = 73 - 61 = 12 \text{ kg.}$$

The Characteristics (Features) of the Range

1. It is easily understood and does not require lengthy calculation.
2. We use it mainly when we are interested in quick but not accurate picture of the variation in a set of data.
3. It is a poor measure of dispersion because it makes use of only two extreme numbers in a set of data.
4. The value is easily affected by one large number.

Advantages of the Range

1. It is easy to understand.
2. It is very simple to calculate.
3. It is used in practice as a measure of dispersion in connection with statistical quality control.
4. It is time saving.
5. It helps in the study of variations in Exchange.

Disadvantages of the Range

1. It is a poor measure of dispersion because it makes use of only two extreme numbers in a set of data.
2. The value is easily affected by one large number.
3. It is possible for the ranges of two groups to be the same and yet their dispersions may be different.
4. It is liable to mislead if unrepresentative extreme values occur.

5. It fails totally to show the degree of clustering, e.g. 2,3,3,3,3,5, has the same range as 2,2,2,5,5,5, but in the former group, value cluster much more closely together.

(ii) VARIANCE

Variance is the average of the squared deviation of measures from their mean. The variance is denoted by the symbol S^2 .

Formula used in calculating variance is as follows:

a) For Ungrouped Data

$$S^2 = \frac{\sum (X - \bar{X})^2}{n-1} = \frac{\sum d_i^2}{n}$$

Where

$$d_i = (X_i - \bar{X})$$

Variance is operated by squaring, hence is expressed in square units instead of its original unit. If the measure was in kg, the answer comes out in kg^2 . This is a disadvantage of variance as the unit of measurement is squared.

Example: Assuming we have a set of observation: 5, 3, 2, 6 and 4.

Solution

$$\text{Step 1: } \sum x = 5 + 3 + 2 + 6 + 4 = 20$$

$$n = 5$$

$$\bar{X} = \frac{\sum x}{n} = \frac{20}{5} = 4$$

$$\text{Step 2: } (X - \bar{X})^2$$

For the 5 observations:

5	$(X - \bar{X})^2 = 5 - 4 = (1)^2 = 1$
3	$(X - \bar{X})^2 = 3 - 4 = (-1)^2 = 1$
2	$(X - \bar{X})^2 = 2 - 4 = (-2)^2 = 4$
6	$(X - \bar{X})^2 = 6 - 4 = (2)^2 = 4$
4	$(X - \bar{X})^2 = 4 - 4 = (0)^2 = 0$
$10 = \sum (x - \bar{x})^2$	

Step 3: Substitute into formula:

$$\frac{10}{5-1} = \frac{10}{4} = 2.5^2$$

- (b). For the **variance of grouped data** any of these formulae can be used.

$$\text{Variance } (S^2) = \frac{\sum f(X - \bar{X})^2}{\sum f}$$

OR

$$S^2 = \frac{\sum fX^2}{\sum f} - \bar{X}^2$$

Where:

X = midpoint

f = frequency

$$\bar{X} = \text{mean} = \frac{\sum fX}{\sum f}$$

Using the aforementioned set of data on the age bearing woman

Age (yrs)	F	X	fx	$F(X - \bar{X})^2$
15 - 19	5	17	5 (17) 85	$5 (17 - 27)^2 = 500$
20 - 24	10	22	10 (22) 220	$10 (22 - 27)^2 = 250$
25 - 29	15	27	15 (27) 405	$15 (27 - 27)^2 = 0$
30 - 34	8	32	8 (32) 256	$8 (32 - 27)^2 = 200$
35 - 39	3	37	3 (37) 111	$3 (37 - 27)^2 = 300$
40 - 44	2	42	2 (42) 84	$2 (42 - 27)^2 = 450$
	$\sum f = 43$		$\sum f = 1161$	1700

Step 1: Get the mean using $\bar{X} = \frac{\sum fX}{\sum f}$

$$\bar{X} = \frac{1161}{43} = 27$$

Step 2: Calculate the x column (midpoint) by adding the lower limit to the upper limit and divide by 2

Step 3: Calculate variance (S^2) by substituting into formula

$$S^2 = \frac{1700}{43} = 39.5$$

NOTE: The disadvantage of variance is that the answer is in square root.

(ii) STANDARD DEVIATION

Standard deviation is the square root of the mean of the squared deviation from the Arithmetic mean of the distribution. Standard deviation is sometimes called the root square mean deviation'. It is the most important measure of dispersion. It must be used whenever the measure of dispersion is needed for further computation. Standard deviation can simply be defined as the square root of variance.

(i) Standard Deviation (Ungrouped Data)

$$\text{This is defined as } SD = \sqrt{\frac{\sum (X - \bar{X})^2}{n}}$$

Sample: Calculate the standard deviation for the following sets of numbers 2, 3, 6, 9 and 15

Solution

Step 1: Write down the formula: $SA = \sqrt{\frac{\sum (X - \bar{X})^2}{n}}$

Step 2: Compute the mean $(\bar{X}) = \frac{2+3+6+9+5}{5} = \frac{25}{5} = 5$

Step 3: Create the table

Observations (X)	Deviation from mean d_1 $= (X - \bar{X})^2$	Deviation from mean $d_2 = (X - \bar{X})^2$
2	$2 - 5 = -3$	9
3	$3 - 5 = -2$	4
6	$6 - 5 = 1$	1
9	$9 - 5 = 4$	16
5	$5 - 5 = 0$	0
25		$30 \sum (X - \bar{X})^2$

$$\text{Variance} = S^2 = \frac{\sum (X - \bar{X})^2}{n} = \frac{30}{5} = 6$$

$$S^2 = 6$$

$$\therefore S.D. \sqrt{S^2} = \sqrt{6} = 2.4$$

b) Standard Deviation (Grouped Data)

This standard deviation is found by:

Squaring the deviations of individual values from the Arithmetic mean.

Adding the squared deviations together.

Dividing this sum by the number of items in the distribution.

Taking the square root of the value as stated above.

Example: (Grouped data)

Below is a table with weights of 20 infants

Wt. of infants (kg)	No of observed. (f)	Mid point (x)	Xf	$d_1 = X - \bar{X}$	$(X - \bar{X})^2$ $f = d_2 f$
1.5 - 3.5	2	2.5	5	- 4.1	33.6
3.5 - 5.5	5	4.5	22.5	-2.1	22.05
5.5 - 7.5	6	6.5	39	- 0.1	0.06
7.5 - 9.5	4	8.5	34	1.9	14.4
9.5 - 11.5	3	10.5	31.5	3.9	45.6
	20		132	0.5	115.71

Calculate the mean

$$\frac{\sum Xf}{\sum f} = \frac{132}{20} = 6.6$$

$$\sum (X - \bar{X})^2 = 115.71; \quad \sum f = 20$$

$$S^2 = \frac{\sum (X - \bar{X})^2}{f} = \frac{115.71}{20} = 5.79$$

$$\therefore SD = \sqrt{5.79} = 2.41$$

Features of Standard Deviation

1. Its calculation includes every item in the distribution.
2. It is amenable to further mathematical calculation.
3. It is the best measure of dispersion and most frequently used.
4. The lower the standard deviation, the more clustered observation.

Advantages of Standard Deviation

1. It reflects the dispersion of scores so that the variability of the different distribution can be compared.
2. It permits the precise examination of scores within a distribution.
3. It is a member of the mathematical system which is amenable to more advanced statistical analysis.
4. It uses all the values in the distribution.

Disadvantages of Standard Deviation

1. It is difficult to understand.

2. It gives more than proportional weight to extreme values because it squares the deviations.
3. It is tedious to compute.

(iv) COEFFICIENT OF VARIATION

The standard deviation provides considerable information about the absolute dispersion of a given distribution, the higher the standard deviation, the greater the amount of scatter. On occasions, it is useful to have a measure of the relative dispersion of a distribution particularly if distributions are being compared. The measure which provides this relative view is the coefficient of variation or, as it is sometimes known, the coefficient of dispersion. This is simply the ratio of the standard deviation of a distribution to the mean of the distribution, expressed as a percentage.

Coefficient of variation =

$$\frac{SD}{\bar{X}} \times 100 \quad \text{OR} \quad \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

Example: Given the distribution X with S.D = 6.2 and mean = 45 and distribution Y with S.D = 2.6 and mean = 18.5. Calculate the coefficient of variation and which of the distribution is relatively more variable.

Solution

Coefficient of variation

$$A. = \frac{6.2}{45} \times \frac{100}{1} = 13.78\%$$

$$B. = \frac{2.6}{18.5} \times \frac{100}{1} = 14.05\%$$

Therefore, distribution B is relatively more variable.

Advantages of Coefficient of variation

- (1) It gives idea of the variability of two sets of figures.
- (2) It is easy to calculate.

Disadvantages of Coefficient of variation

- (1) They relate to sets of figures of quite different orders of size.
- (2) They are in different units.

(v) QUANTILES

Quantiles are the types of averages that divide a distribution into 4 equal parts. There are three quantiles in a set of any distribution (Q1, Q2 and Q3)

	Q1
	Q2
	Q3

Since we know how to calculate the median, which is the second quartile (Q2), we shall be concerned with only the calculation of Q1 and Q3.

The formula for calculating the first quartile Q1 is:

$$Q1 = L_1 + \frac{N - (\sum f)b) c}{4 FQ1}$$

Where: L1 = True lower limit of the first quartile class,

N = Sum of frequencies; c = class interval.

$(\sum f)b$ = Sum of all frequencies below the Q1 class.

FQ1 = frequency of the Q1 class.

Similarly the formula for Q3 is:

$$Q3 = L_1 + \frac{(3N - (\sum f) b) c}{4 FQ3}$$

Where: FQ3 = frequency of 3rd quartile class.

Example: Given the following data of distribution of marks scored by 120 students in Anatomy and Physiology.

Marks	30-39	40-49	50-59	60-69	70-79	80-89	90-99
Frequency	1	3	11	21	43	32	9

Calculate (a) the first Quartile (Q1) and (b) Third Quartile (Q3).

Solution: Determine the Quartile classes and re-write the table and create Cumulative Frequency.

$$\text{First Quartile class} = \frac{N}{4} = \frac{120}{4} = 30$$

$$\text{Third Quartile class} = \frac{3N}{4} = \frac{120}{4} \times 3 = 90$$

Create table using true upper/lower class limits.

Marks	Frequency	Cumulative frequency
29.5 - 39.5	1	1
39.5 - 49.5	3	4
49.5 - 59.5	11	15
59.5 - 69.5	21	36 1 st Quartile Class
69.5 - 79.5	43	79
79.5 - 89.5	32	111 3 rd Quartile Class
89.5 - 99.5	9	120
Total	120	

NOTE: To determine the first quartile class, 30 falls between 15 and 36 of the cumulative frequency, thus 59.5 - 69.5 is the Q1 class. The quartile class, 90, falls between 79 and 111, therefore, 79.5 - 89.5 is taken as the third quartile class.

Step 3: Substitute into the formula:

$$Q1 = 59.5 + \frac{(120 - 111) \times 10}{21} = 59.5 + \frac{(9 - 1) \times 10}{21}$$

$$Q1 = 59.5 + \frac{(19) \times 10}{21} = 59.5 + \frac{190}{21} = 59.5 + 9.05 = 68.55$$

(b) Q3 (the third quartile)

$$Q3 = 79.5 + \frac{(111 - 79) \times 10}{32} = 79.5 + \frac{(32) \times 10}{32} = 79.5 + 10 = 89.5$$

(vi) RELATED MEASURES OF VARIATION

The Interquartile and Semi-Interquartile Range

In some special problems, it may be convenient to use a method of variation, which is easy to find but yet more informative and stable than range. Such a measure of dispersion is the Inter-quartile range. Inter-quartile range is defined as the difference between the first and third quartile = $Q_3 - Q_1$

(a) Quartile Range

Example: Suppose we are given frequency distribution where the calculated data first and third quartiles are $Q_1 = 66.64$ and $Q_3 = 82.94$.

Find the Inter-quartile Range

$$\begin{aligned} \text{Solution: Inter-quartile Range} &= Q_3 - Q_1 \\ &= 82.94 - 66.64 = 16.30 \end{aligned}$$

Advantages of Inter-quartile Range

1. It takes care of 50% middle variation of the data.
2. It is unaffected by large and small values as in the case of the range.

Disadvantages of Inter-quartile Range

1. It does not by itself allow the exact interpretation of any item within the distribution.
2. It does not lend itself to further mathematical treatments that are required for inferential statistics.

(b) Semi-Interquartile Range (Quartile Deviation)

This is defined simply as $\frac{Q_3 - Q_1}{2}$

This measure of dispersion gives the average amount by which the two quartiles Q_3 and Q_1 differ from the median.

This is because the median is also the second quartile.

Example: A distribution has first quartile = 66.64 and third quartile 82.94

Solution: Quartile Deviation = $\frac{Q_3 - Q_1}{2}$

$$= \frac{82.94 - 66.64}{2} = \frac{16.3}{2} = 8.15$$

Advantages of Quartile Deviation

1. It is simple to calculate once first and third quartiles are known.
2. Simple to understand.

Disadvantages of Quartile Deviation

1. It fails to take into account all the values in the distribution.
2. It shows no true indication of the degree of clustering.



CHAPTER 6

TEST OF HYPOTHESIS

Statistical tools used for data analysis is either parametric or non-parametric.

Parametric or non-parametric tools or techniques may be adopted depending on the objective(s) of the study and type of sample. The use of appropriate test statistic is imperative

Test of hypothesis could be referred to as a test of significance. A hypothesis is a statement whose identity is to be verified. E.g. If the average systolic pressure of lawyers between the ages of 45 - 50 years is 135mmhg while the sample mean of the systolic pressure of the lawyers have changed from 135mmhg to 130mmhg, in order to arrive at a conclusion, one has to subject the sample mean to test of significance to confirm that the blood pressure of the concerned group has reduced from 135mmhg to 130mmhg.

Null Hypothesis

When there is no difference between the population mean and the sample mean, the hypothesis is said to be null hypothesis; represented as:

$$H_0 : X = \mu : (\text{mean})$$

Alternate Hypothesis

Where there is a difference between the population mean and the sample mean, the hypothesis is said to be alternate hypothesis.

Some authors refer to it as directional. The difference between the two means is not by chance. It is represented as:

$$H_1 : X < \mu$$

Or

$$H_1 : X > \mu$$

If one accepts the null hypothesis (H_0), the alternate hypothesis (H_1) is rejected. On the other hand, if one accepts the alternate hypothesis, this shows there is a significant difference and as such the null hypothesis is rejected. Two types of errors are associated with rejection or acceptance of a hypothesis, namely: type I and type II errors.

Type I Error

This is the type of error which is said to have been committed if one rejects a true hypothesis and this is represented by α (alpha).

Type II Error

This type of error is said to occur when there is the probability of one accepting a wrong hypothesis i.e. accepting a hypothesis, which should have been rejected. It is represented by β (beta).

Level of Significance

When hypothesis is being tested, the level of significance is tested. A level of significance is the probability of committing a type I error or maximum probability at which a hypothesis is rejected. Type I error which is represented by ' α ' is the same thing as the p value, the common p values in most studies are 5% and 1%. This means that if

anyone formulates a hypothesis and wants to test it with a 5% error confidence, it means that it has 95% confidence level. i.e.

$$H_0 : X = \mu \text{ 95\% confidence}$$

If the test has 1% error confidence, it means that it has 99% confidence level. (The probable error in the result is only 1%). p value of 0.05 is more common in use.

When testing the level of significance, it could be one tailed test or two tailed test.

One Tailed Test

A test is said to be one tailed if it considers one side of the normal distribution curve. It is a significance that it is a directional test; either to the left or to the right.

In one tailed test;

$$H_1 : X < \mu \text{ or } > \mu$$

Two Tailed Test

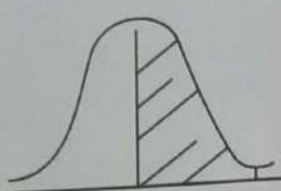
A test is said to be two tailed when two sides of the distribution curve are considered. It is a non-directional test; 'Not equal to' which indicates a two-tailed test. It involves comparing two populations.

e.g., Population A and B

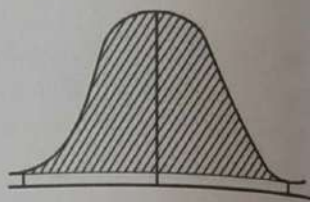
A B

X_1 X_2

$$H_1 : X_1 \neq X_2$$



One Tailed



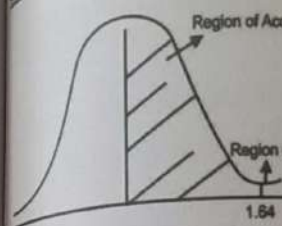
Two Tailed

The rejection and acceptance of hypothesis depends on the critical value. Critical values are values of the test statistic that separates the acceptance region from the rejection region. In one tailed, test, the critical value is 1.64, which is the rejection area.
 $Z = 1 - 0.05 = .95 = 1.64$ (from table)

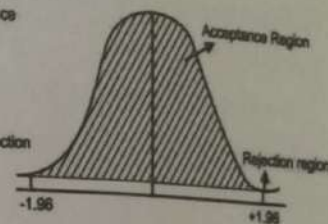
In a two-tailed test, the critical value is 1.96, which is the rejection area.

$$\frac{0.05}{2} = 0.025$$

$$Z = 1 - 0.025 = 0.975 = 1.96 \text{ (from table)}$$



One Tailed



Two Tailed

When test of hypothesis is being employed, it is not on total observation of a population but the sample population.

STEPS

1. Let $X_1, X_2, \dots, X_n$ represent the set of observations
2. Compute the sample mean, where:

$$\bar{X} = \frac{\sum X}{n} = \mu$$
3. Decide on the appropriate test statistic to use. E.g., is it the Z score, the t test or chi-square test?
4. Choose the level of significance or the P value. The two most common P value in use are P. 0.05 and P 0.01. 0.05 is testing at a confidence level of 95% or error of 5% while 0.01 is testing at a confidence level of 99% or error of 1%.
5. Determine the critical value. This is dependent on whether the hypothesis involves a one tailed or two tailed test.

6. Calculate the value of the statistics using the formula of your choice.
7. Compare the calculated value with what you have in the table. If calculated value is greater than the critical value (positive) reject the null hypothesis. Similarly, if the calculated value is less than the critical value, reject the alternative hypothesis and accept the null hypothesis. If one accepts the hypothesis, one is saying that there is no significant difference. i. e. ($P > 0.05$). If one rejects the null hypothesis, one is saying there is significant difference. i. e. $p < 0.05$. (Significant difference).

Whenever result is being interpreted, level of significance and p-value should be compared; Whenever p-value is greater than α -value, null hypothesis is accepted and then concluded appropriately. Whenever p-value is less than α -value, null hypothesis is rejected and then conclude appropriately. Rejecting null hypothesis, may lead to further test, e.g. in Regression and ANOVA.

PARAMETRIC TESTS

Some simple tests are employed in the test of hypothesis known as inferential statistics. Statistical calculation is done using approved formulae to find out if there is any significant difference in the findings. The simple tests used are as follows:

A NORMAL DISTRIBUTION

Usually, for large sample size greater or equal to 30, the normal distribution is assumed unless otherwise stated. Hence, for testing

hypothesis involving the use of normal distribution we use the standardized value 'z' called the z score.

$$Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{N}}}$$

Where:

$\bar{X}$ = Sample mean

S = Sample standard derivation

N = size of the sample

μ = population mean when given

QUESTION 1

The manufacturing process used by Fenice Pharmaceutical chemist, Nkpor during the period 2015 to 2017 yielded an average output of 100 units per hour with standard deviation of 10. A new and more expensive machine for making the product has just been put to the market and the manufacturer claimed that in general, it introduces 150 units per hour on the average with standard deviation of 8 units. The company bought 35 of these machines for experimentation and found that the average output was 160 units per hour with standard deviation of 8 units.

Requirement: To determine whether the mean output of these 35 machines being greater than 150 units per hour at 5% level of significance is due to chance.

ANSWER:

Step 1: formulate the hypothesis

$H_0 = U_0 = 150$ = null hypothesis

$H_n = U_A = 150$ = alternative hypothesis

Step 2: Select the testing statistic

This is normal distribution problem since the size of the sample (n) is greater than 30.

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{N}}}$$

Step 3: Calculate the Z - value:

$$Z = \frac{160 - 150}{\frac{8}{\sqrt{150}}} = \frac{10}{1.3} = 7.41$$

Step 4: Determine the value of significance, which in this case is 5% level of significance or 95% confidence level. Since it is a one-tailed test, we go to table of area under normal curve to find the Z- value corresponding to $0.5 - 0.05 = 0.45$ which is 1.64.

Step 5: Decision and conclusion: Hence, the decision rule is that S calculated is greater than Z from the table, we reject the null hypothesis that the average output of the new machine is 150 units and accept the alternative hypothesis that the average output of the machine is greater than 150 units and the occurrence is not by chance.

B. THE STUDENT 'T' DISTRIBUTION (Dependent & Independent)

The 't' distribution is used for significance testing when the sample size is less than 30. The number of degrees of freedom, df, is $n - 1$ or for two samples $n_1 + n_2 - 2$ (two). However, if the population standard deviation is known then normal distribution can be used. The 't' distribution is required when we have the estimate σ from s where σ = population standard deviation and s = sample deviation.

't' distribution =
$$\frac{\bar{X} - \mu}{\frac{s}{\sqrt{N-1}}}$$
 where s is an estimate of σ

Where $\bar{x}$ = sample mean
 μ = population mean

n = size of the sample
 s = standard deviation

QUESTION 2

It was observed in 2015 that the mean expenditure of maintaining a cellular phone among a population of cellular phone owners in a school is of the average of N30 per day. The sample of 17 cellular phone owners were selected at random and observed that the mean expenditure on daily phone maintenance was N34 with standard deviation of N8. Does the result show that the mean cellular phone maintenance is greater than normal at 5% level of significance?

ANSWER: Step 1: write down the formula for $\mu = 30$:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$$

$n = 17; \bar{x} = 34; \mu = 30 \text{ and } s = 8$

Step 2: The test: $H_0: \mu = N30; H_A: \mu > N30$

Step 3: Substitute into the formula:

$$t = \frac{34 - 30}{\frac{8}{\sqrt{17-1}}} = \frac{4}{\frac{8}{\sqrt{4}}} = \frac{4}{2} = 2$$

Step 4: Look for the degree of freedom from table with degrees of freedom:

$$17 - 1 = 16; t_{0.095} = 1.75$$

Step 5: Decision and conclusion:

Since calculated $t = 2$ is greater than $t = 1.75$, we reject the null hypothesis that the average expenditure is equal to N30 and accept the alternative that the average expenditure is greater than N30.

C. THE Chi-square (X^2) Distribution:

The X^2 test is used when it is wished to compare an actual observed distribution with a hypothesized or expected distribution. It is often referred to as a 'goodness fit' test, test of independence and test for association. In this test, null hypothesis is usually formulated which says that the two things under consideration are independent of one another. It is a distribution which all that is observed is in the positive because even when a negative value is obtained, when squared it becomes a positive value.

The Chi-square is defined as:

$$X^2 = \sum \frac{(O-E)^2}{E}$$

Where: O = the observed frequency of any value

E = the expected frequency of any value

The X^2 (chi-square) value obtained from the formula is compared with the value from X^2 table. The distribution of X^2 shows the various degrees of freedom (df) and the cut off values at the two most common significance levels of 5% and 1%

QUESTION 3

Given the following data on the observed frequency of tossing a die 110 times.

Face	1	2	3	4	5	6	Total
Observed Frequency	20	17	10	23	24	16	110

Required: Test the hypothesis that the die is unbiased at 95% confidence level.

ANSWER

Step 1: Write down the formula:

$$X^2 = \sum \frac{(O-E)^2}{E}$$

Step 2: Determine E: from the table we expect each face to occur with a probability:

$$1/6 \times 110 = 18.33$$

Step 3: Create the table:

Face	Observed Data (O)	(o-e)	(o-e) ²	$\frac{(o-e)^2}{e}$
1-	20	20 - 18 = 2	4	0.1
2-	17	17 - 18 = -1	1	0.06
3-	10	10 - 18 = -8	64	3.6
4-	23	23 - 18 = 5	25	1.4
5-	24	24 - 18 = 6	36	2.0
6-	16	16 - 18 = -2	4	0.2

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$$\frac{(o-e)^2}{e} = 7.4$$

Step 4: Find the degree of freedom (df):

$$df = N - 1 = 6 - 1 = 5$$

$$= 0.95 = 11.1 \text{ (level of significance)}$$

Step 5: Decision and Conclusion:

Since $X^2 = 7.4$ and is less than $X^2 = 11.1$, we accept the null hypothesis that the die is unbiased.

Note:

Independent t-test (Regular t-test)-Is for comparison of significant difference between the mean of two independent groups.

Dependent t-test (Matching -t test) -Is used to compare two group of subjects pre-tested and post-tested or two groups matched against each other.

Degree of Freedom: df

Independent t-test = $(N_1 + N_2 - 2)$ or total no of respondents minus 2
Dependent t-test = $(N_1 + N_2 - 1)$ or total no of respondents minus 1

NON -PARAMETRIC TECHNIQUE

Here are some examples of non- parametric tests that can be used as equivalent:

PROPER APPLICATION OR USE OF STATISTICAL TEST

Appropriate application of statistical tests in the test of hypothesis is important for validity of the result interpreted. The researcher or the statistician must know when to use parametric test, non-parametric or both.

One Sample data test

If data is drawn from normally distributed data, Z -test is to be used if $n > 30$. Otherwise t-test could be used. If the condition of parametric test is not met, non parametric equivalent like one-sample sign test or Wilcoxon's signed rank test can be used.

Two Independent Samples test

If data is drawn from independent and normally distributed data, t-test is to be used.
If condition of parametric test is not met, then non parametric equivalent, Mann-Whitney test is to be used.

Two Matched Samples Test

Appropriate test is paired t-test which is parametric. The non parametric test alternative is Wilcoxon's signed rank test.

Three or more Independent Samples test

The appropriate test is one way ANOVA which is a parametric test. If the condition of parametric test is not met, non parametric test equivalence is Kruskal -Wallis test.

Three or more Matched or related Samples test

Parametrically, appropriate test is two way ANOVA. The non parametric test equivalent to two way ANOVA is Friedman test.

Test for Linear Relationship

Parametrically, appropriate test statistic is Pearson's product-moment Correlation Coefficient.

Non parametric tests equivalence are Spearman's rank or Kendall's tau Correlation Coefficients.

Test for equality of two population variances

F-Test is the appropriate parametric test statistic while the non parametric tests equivalence is Ansorri-Bradley test or Moses test..

Fitting Linear Model

Parametrically, appropriate test statistic is Simple Linear or Multiple linear Regression Models.

Non parametric test equivalence is Theil's Method of linear Regression Models.

Test for equality of several population variances

One can use any of following parametric tests. They are: Bartley test, Cochran-C test, Fisher's F-max test or Levene's test while non parametrically, Squared rank test is to be applied.

Use of Chi-square

Chi -Square is used when we wish to test:

- More than two populations proportion
- For independency of sample data
- Goodness of fit of particular distribution in a given data

CHAPTER 7

PRESENTATION OF DATA

Presentation means organization of data i.e. re-arrangement in order to facilitate communication.

Aims of Presentation:

1. To rearrange original data in order to answer research questions.
2. Primary and secondary data need to be arranged and presented in some ways before information contained in the data can be interpreted.
3. Tabulation help to facilitate understanding of numerical data.

Basic Principles of Data Presentation

1. Data should be presented in a clear and good manner.
2. The information should be presented as attractive as possible.

The Rules To Be Observed in Data Presentation:

- a. There must be a statement in relation to the source of information.
- b. Every chart or table must have a title.
- c. The scales used must be stated.
- d. Every chart or table must have one unified set of relationship.
- e. Heading for both vertical and horizontal columns must be clearly worded.

The subject matter must be clearly presented.

The most important points in the data must be clarified.

Purpose of the presentation must be borne in mind.

The amount of detail and accuracy has to be put into consideration.

VARIOUS FORMS OF PRESENTATION

The two main ways of presentation of data are tabular and graphical (diagrams, pictures) methods.

A) TABULAR METHOD

The information/data are presented in form of tables, otherwise it remains a raw data. The points to note when tabulation is being employed are as follows:

Points to Note:

- i. Table must be set up correctly
- ii. It must be identified fully and most important points must be clarified.
- iii. Table must contain appropriate information.
- iv. The purpose of presentation must be considered.
- v. The amount of detail required must also be considered.
- vi. It has to be simple and unambiguous.
- vii. Tabulation should help to facilitate understanding of complex numerical data.

Classification of Table:

Table can generally be divided into:

- i. Informative table, which provides statistical record.

- ii. Reference table containing summarized information.
- iii. Complex table showing number of columns, which are inter-related.

Component or Parts of Tables

The table has various components, which include:

i. The table number

ii. Title.

iii. Caption (Column sub title)

iv. Stub (Row sub-title).

v. Source.

- The title explains the subject matter of the table.

- The caption is the column sub-title that gives explanation to the title along the vertical row. This is the frequency column i.e., along the Y-axis.

- The stub (Row sub-title), gives explanation of the table along horizontal row. This column is mainly for the class interval i. e., X-axis.

- The source explains the place or site where the information is collected especially with secondly information. Source is placed at the bottom of the table.

Methods of Tabular Presentation:

(i) Frequency Distribution Table:

It is the simplest form of data presentation. It shows the number of occurrences of each variable, e. g. age can be used here as the variable. Assuming 38 people aged between 0 - 25 years are recruited in a study, the table below shows their age distribution.

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AGE (years)	Frequency	Percentage
0 - 4	5	13.2%
5 - 9	10	26.3%
10 - 14	15	29.5%
15 - 19	5	13.2%
20 and above	3	17.8%
Total	38	100%

(ii) Ratio

Ratio is used to compare two events or two variables. It shows the magnitude of one event over the other e.g., male and female. If a total of 800 people in Umudim participated in a study, 200 males and 600 females, ratio summarizes it as 200 (M): 600 (F) i.e. 1:3.

(iii) Proportion i. e. %

This is a type of ratio but has the numerator added to the denominator. It is usually expressed in percentage e.g., using above data, the proportion of men in the study will be

$$\frac{200}{200 + 600} \times 100 = 23\%$$

$$\frac{200}{200 + 600} = \frac{1}{4}$$

(iv) Averages

This will include arithmetic mean, mode and median. These are called measures of central tendency. Mean is the sum of all the items in a set of data, divided by the number of items in the set of data while median is the middle observation in a set of data that has been arranged in a particular order of magnitude. Mode is the most

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occurring frequency in the distribution. Any of these can be used to summarize data or findings. (See previous chapter for examples)

v. Rate

This is the number of events in a particular time or period over the population at risk.

K = Constant. K can represent 100, 1000 etc. Most commonly used is one thousand (1000). Examples of rate include incident rate, prevalence rate, crude death rate, mortality rate etc.

B. DIAGRAMMATIC PRESENTATION OR PICTURAL (GRAPHS)

This helps to know at a glance the characteristics of the study. Various forms of graphs are:

- * Line chart
- * Bar chart
 - i. Simple bar chart
 - ii. Compound bar chart
 - iii. Component bar chart
- * Pie chart
- * Histogram
- * Frequency polygon
- * Relative frequency distribution

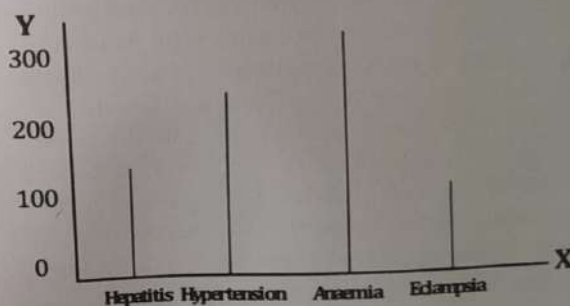
Cumulative frequency distribution (Ogive)

1. LINE CHART

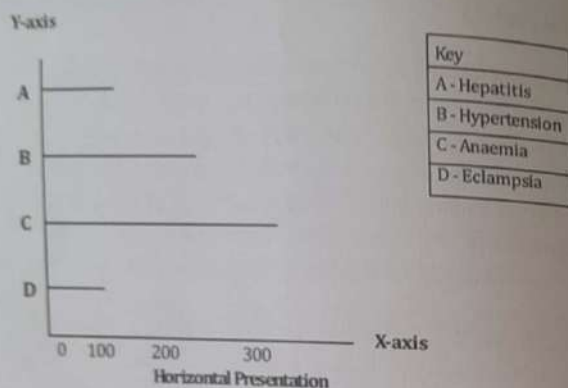
Line chart is used to represent either a qualitative or discrete quantitative variable (variable that is uncountable and integral) using single lines. Lines can be arranged vertically or horizontally, although the vertical arrangement is usually preferred. The lengths of the lines are proportional to the magnitude of the observed value. Example of records of selected diseases in one year (2020) are as follows:

Disease:	Hepatitis	Hypertension	Anaemia	Eclampsia
Person	150	230	300	100

Vertical Presentation



Horizontal Presentation



2. BAR CHART

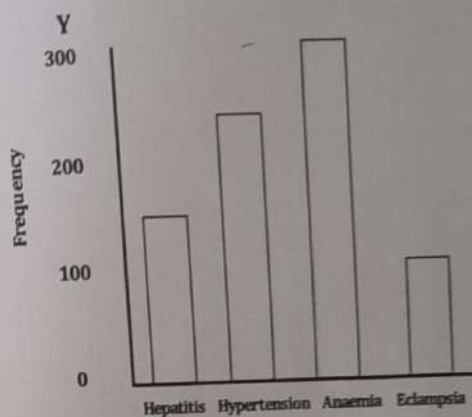
The bar charts are bars (rectangles) which are of equal width and whose lengths are proportional to the quantities (or frequencies) they represent but which are being separated by equal intervals. The bars are used to represent the qualitative and discrete quantitative variables. Representation is same as the line chart. Vertical arrangement is preferred. Lengths of bar are proportional to the magnitude. Widths are not important but gap should be created between one bar and another. Bar have bolder picture. The difference between line and bar charts are that line charts are converted to rectangular blocks. Both do the same work.

There are three (3) types of bar charts, namely:

- *simple
- *Compound
- *Component

*Simple Bar Chart

The above data on diseases can be represented using simple bar chart.

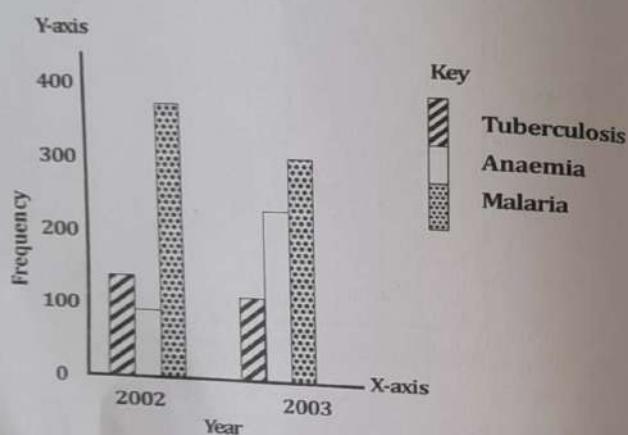


*Compound Bar Chart

It is used for representation of qualitative and discrete variable. Bars are usually used to represent observations made over a period of time. The bars are in groups. A gap is usually created between periods and within a period of time.

Example: If the data collected on the type of diseases of various patients in a particular hospital within two years is as follows:

Diseases	2002	2003
Tuberculosis	150	100
Anaemia	90	250
Malaria	350	300



* Component Bar Chart

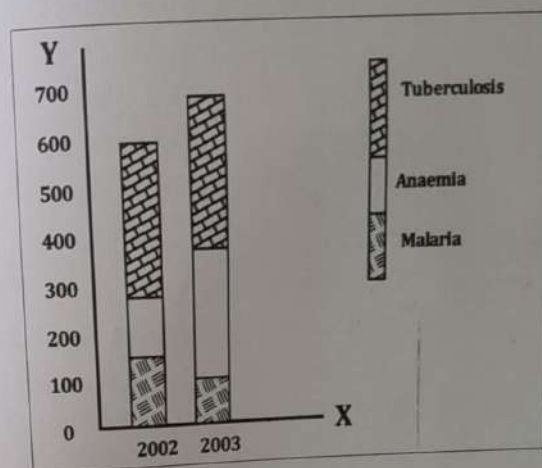
In this type, the lengths of the bars are proportionate to the magnitude of the sum of the data. Only one bar is used at a time.

Using the above data:

For 2002, it is $150 + 350 + 90 = 590$

For 2003, it is $100 + 250 + 300 = 650$

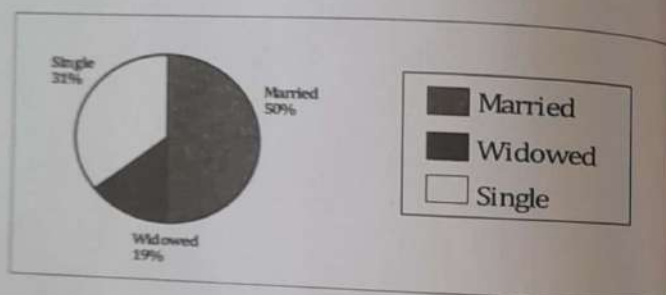
Diseases	2002	2003
Tuberculosis	150	100
Anaemia	90	250
Malaria	<u>350</u>	<u>300</u>
	590	650



3. PIE CHART

This is circular representation of a discrete quantitative and qualitative variable. The circle is usually partitioned into sectors with the area of the sectors being proportional to the magnitude of the items. The original observation or frequency is converted to their degree equivalent. Example if the marital status of 320 women in a study is represented thus:

Marital Status	Frequency	Degree		
Single	100	$\frac{100}{320}$	$\times 360$	$= 112.5^\circ$ i.e.
31%			1	
Married	160	$\frac{160}{320}$	$\times 360$	$= 180^\circ$ i.e.
50%			1	
Widow	60	$\frac{60}{320}$	$\times 360$	$= 67.5^\circ$ i.e.
19%			1	



Pie Chart representing the marital status of the women in the study.

4. HISTOGRAM

Histogram is a graphical representation of continuous quantitative variable (data). They are overlapping rectangular blocks in which the length of the rectangle is proportional to the magnitude of the frequencies. The widths are equally represented and are joined to the corresponding class. Frequencies are plotted on the vertical axis and the class intervals or the values on the horizontal axis. The

heights of the histogram are determined by the corresponding class frequency.

Steps involved in plotting a histogram are as follows:

- Any array of data must be arranged in ascending order.
- Determine range.
- Determine class interval.
- Re-arrange table using the true lower and upper limits and draw histogram.

Example:

Assuming the ages of 20 employees in Fenice Pharmaceuticals chemist is as follows:

50, 57, 66, 26, 46, 59, 35, 43, 75, 64,
50, 61, 52, 69, 48, 45, 42, 49, 54, 67.

Solution:

Step 1: Re-arrange the data in ascending order:

26, 35, 42, 43, 45, 48, 49, 50, 52, 54,
59, 61, 64, 66, 67, 69, 75.

Step 2: Determine the range i.e. the difference between largest and smallest number.

$$= 75 - 26 = 49 \text{ (Maximum - Minimum)}$$

This is important in determining the number of classes we are to group the data.

Classes varying between 5 and 20 are used.

To determine number of classes that will be created, Sturge's Rule is employed.

$$\text{Sturge's rule} = 1 + 3.322 \log n$$

(To work this out, press n, then Log.)

Example: If $n = 100$ i.e. number of respondents,

$$1 + 3.322 \log 100 = 1 + 3.322 \times 2 = 7.644 = 8 \text{ groups or classes}$$

Class interval is also worked out.

Class interval = maximum number *minus* minimum number *all over* number of groups

$$\text{i.e. } \frac{\text{max} - \text{min}}{\text{no. of groups}}$$

For the above data, assuming we use 5 classes at intervals of 10. Thus, we count 10 starting from the lowest number, which is 26.

Ages	Counts/tally	Frequency
26 - 35	II	2
36 - 45	III	3
46 - 55	IIII II	7
56 - 65	IIII	4
66 - 75	IIII	4
		$\Sigma f = 20$

Note:

(i) Σf is summation of frequencies i.e. total frequency.

(ii) Frequency is the rate of occurrence or the number of students whose scores fall within that range.

(iii) The above table uses equal class intervals. At times, unequal or open ended class intervals are used.

Step 3: Rewrite the table paying attention to true lower and upper limits. To obtain these limits, we look for the difference between any two classes of the distribution, divide the difference by 2 and add the answer obtained to the number by the right and subtract the number from the number by the left.

Example: The difference between 1st and 2nd Classes = $36 - 35 = 1$

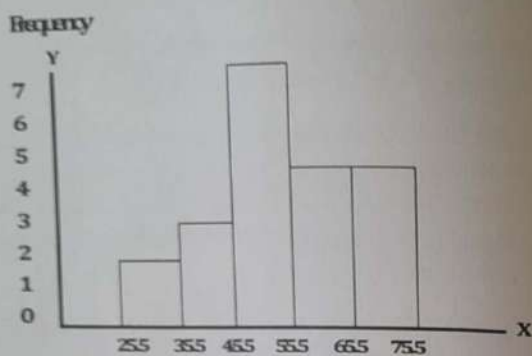
The difference, 1 is divided by 2 = 0.5

The numbers to the left of the distribution are 26, 36, 46, 56 and 66. When 0.5 is subtracted from these numbers, we have 25.5, 35.5, 45.5, 55.5 and 65.5. The number to the right of the distribution are 35, 45, 55, 65 and 75. Then we add 0.5 to these numbers, we obtain 35.5, 45.5, 55.5, 65.5 and 75.5.

Step 4: Re-arrange the table using the true lower and upper limits and draw the histogram.

Scores	Frequency
25.5 - 35.5	2
35.5 - 45.5	3
45.5 - 55.5	7
55.5 - 65.5	4
65.5 - 75.5	4
	$\Sigma f = 20$

Title: The histogram showing ages of the 20 employees.



5. FREQUENCY POLYGON

Frequency polygon is a line graph of a frequency distribution plotted against mid points of the respective class intervals. The graph is obtained by the mid points at the top of the rectangle of the histogram. To complete the frequency polygon, we usually add one class at both ends of the frequency distribution and creat the true lower and upper limits.

Example: Using the data on some 20 employees of Fenice Pharmaceuticals chemist:

Ages	26 - 35	36 - 45	46 - 55	56 - 65	66 - 75
Frequency	2	3	7	4	4

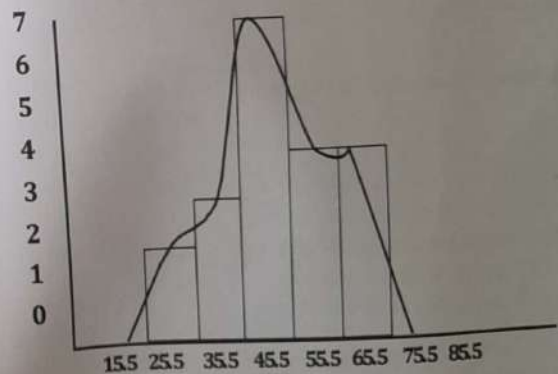
To compute the frequency polygon, we usually add one class to each of the frequency distribution.

Rewrite:

Ages	Frequency
15.5 - 25.5	0
25.5 - 35.5	2
35.5 - 45.5	3
45.5 - 55.5	7
55.5 - 65.5	4
65.5 - 75.5	4
75.5 - 85.5	0

Title: Frequency polygon showing the ages of the 20 employees.

Frequency



6. RELATIVE FREQUENCY DISTRIBUTION

Relative frequency distribution of a class or group is the frequency of the class divided by the total frequency of all classes. If the frequency of each class is replaced by its relative frequency, the resultant distribution is what is called relative frequency distribution. Relative frequency distribution when plotted in form of histogram or polygon is called Relative Frequency Histogram or Relative Frequency Polygon as the case may be.

Example

Assuming that the height of 100 pupils in Community Primary School, Isu-Obisii is thus:

Height/cm	59-61	62-64	65-67	68-70	71-73	Total
No of pupil	10	12	38	25	15	100

To draw a Relative Frequency Histogram

Solutions

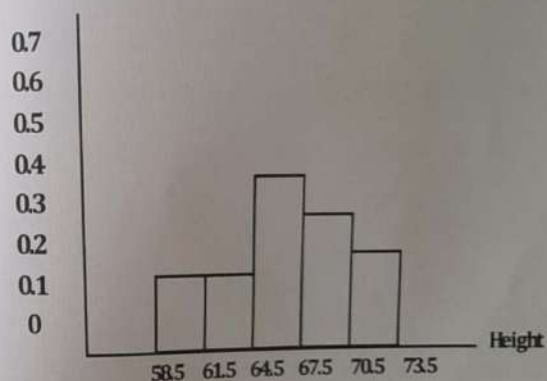
Step 1

- A: For the 1st group, we have $10/100 = 0.10$
 B: For the 2nd group, we have $12/100 = 0.12$
 C: For the 3rd group, we have $38/100 = 0.38$
 D: For the 4th group, we have $25/100 = 0.25$
 E: For the 5th group, we have $15/100 = 0.15$

Step 2: We now re-write the frequency distribution with the true class limits and their corresponding relative frequency, and draw a Relative Frequency Histogram.

Height	Relative Frequency
58.5 - 61.5	0.10
61.5 - 64.5	0.12
64.5 - 67.5	0.38
67.5 - 70.5	0.25
70.5 - 73.5	0.15

Title: Relative Frequency Histogram showing the height of the 100 pupils.

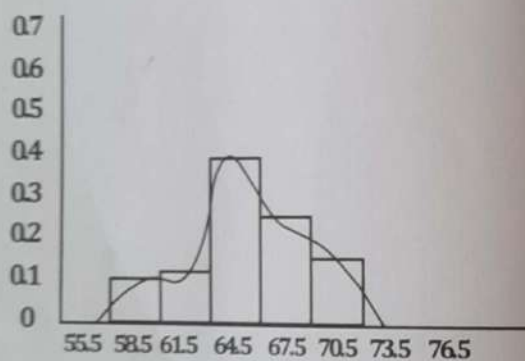


To draw a Relative Frequency Polygon using the same data

The frequency distribution table is re-written creating two lassses, one at the top and the other below. The 2 new classes are assigned the value

Height	Relative Frequency
55.5 - 58.5	0.10
58.5 - 61.5	0.12
61.5 - 64.5	0.38
64.5 - 67.5	0.25
67.5 - 70.5	0.15
70.5 - 73.5	0
73.5 - 76.5	0

Title: Relative Frequency Polygon.



7. CUMULATIVE FREQUENCY DISTRIBUTION (Ogive)

An Ogive is a graph "less than" or "more than" cumulative distribution. A cumulative frequency distribution could be drawn on a "less than" or "more than" basis. Cumulative frequency is calculated by adding the frequencies of classes above it. 1st class is 5, when added to the one below it, 18.

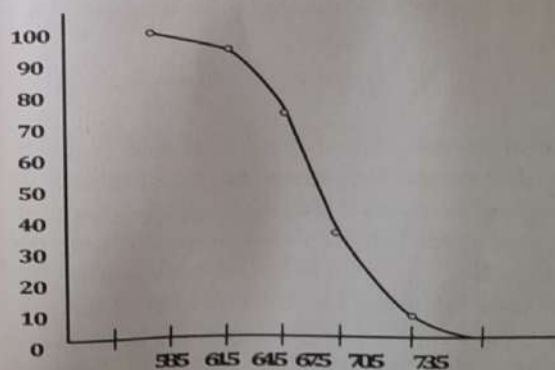
The sum = 23, etc.

Example

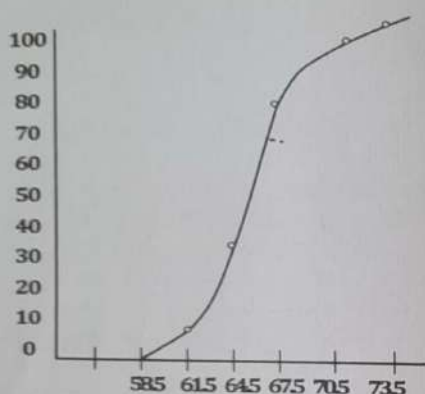
Using the data on the height of the 100 pupil, the table will read thus:

Height	Frequency	Cumulative frequency	
		"Less than"	"More than"
59 - 61	5	5	100
62 - 64	18	23	95
65 - 67	42	63	77
68 - 70	26	91	35
71 - 73	9	100	9

Title: "More than" curve of cumulative frequency distribution.



Title: "Less than" curve of cumulative frequency distribution.



Note: Any of the methods of data presentation could be adopted by the researcher depending on the type of data for presentation, the ability of the researcher or as demanded by the sponsors of the research work.

DISCUSSION

After analyzing and presenting the result in form of tables or graphs, the discussion of findings follows thereafter. The distribution is arranged logically and in sequence. The reaction of the subjects are put down in percentages based on their responses. Only the important findings related to the subject matter are discussed. The researcher compares his findings with the findings of previous researchers on the issue. He states whether or not their findings are in line with his own findings or not. Explanations as to why there are differences or similarities are stated. All work done by previous researchers used in comparison by the researcher must be those of

authors cited in his text. Inference and implications of the findings are stated if need be, in describing the results.

WRITING THE RESEARCH REPORT

After the collection and analysis of data, one will need to communicate to others what he or she have done during the research. It is known as dissertation. Only relevant information is presented in an accurate and concise manner. Some institutions or organizations demand for a report. The report writing normally takes this form.

General Guide for Writing Dissertation (Use format from authority)

1. Title Page

- It must be concise and attractive.
- Must be an eye catcher.
- Should be able to portray what one have studied in the field, in a most effective and efficient manner.

- (i) Approval or Certification page - containing name of supervisor and stating that it is one's original work which is not yet published by any other person.
- (ii) Acknowledgement - write up on acknowledgement of key people that helped one during the research to achieve the desired goal. (i.e thank you page)
- (iii) Dedication Page - this is optional, some dedicate to family members, others to God.

- (iv) Table of content – page numbers of all chapters and contents are stated.
- (v) List of tables and pages where they can be located are shown.
- (vi) List of figures such as; diagrams, histograms, graphs, etc.
- (vii) List of plates; the pictures and their titles and where to find them are also stated. It is optional, though it makes it look more attractive.

(viii) Abstract/Summary

- Just the important findings in the result.
- Provide information on how the research was carried out, i.e. a brief methodology.
- Include the important findings in your result.
- Include the implications of the result.

Abstract/summary should be contained in about 1 or 2 pages and should not be more than 2 pages. A maximum of 200 words is usually ideal.

2. Chapter One

Chapter one then follows and should be written serially, it should also be captivating and appealing to arouse the interest of the readers.

- It should contain a statement of the problem that one worked on, eg., contraceptive practice of Nigerian Women.

- To check population increase.

- Should also include the justification for the study, why the researcher deemed it necessary to carry out the study, like to check population increase.

- There should be theoretical information about the topic. For example, if one is studying a particular disease, one may have to consider the pathophysiology, together with the signs and symptoms and complications/problem of the disease, as well as laboratory investigations and treatment.

- The general aim and objective of the study is also important. One sentence can be used to cover the entire purpose of the study.

- This is followed by the specific objectives; this can be broken into sections in order to make the work easier and explicit. It should not be more than four (4) in number, else it would be confusing and difficult to analyze. It should be clear and concise.

- Limitations of the study – problems or difficulties encountered during the study and whatever one did to eliminate such problems.

3. Chapter Two

This is the Literature Review section.

In this section, one critically reviews previous works or publications related to one's topic and area of study. The researcher also show previous works done on the topic by other researchers as well as quote the researchers, citing

them in the text as approved by one's institution or establishment.

4. Chapter Three

Materials and Method (Methodology).

This is a step-by step method (information) on how the researcher conducted the study. It is a more detailed discussion of the proposal. It usually starts with the study background written in subheadings.

- General Background - Written about the study area, then the demographic characteristics of the people involved in the study i.e their sex, age, occupation, language, climatic, characteristics, etc.

- Research design/method used should be stated. Was it descriptive, cross sectional, quasi experimental, analytical or experimental?

- Define the study population, emphasizing on the sample size formula used to select the sample population and also the method of sampling used.

- Data collection - Did the researcher use any instrument/a questionnaire? Mention the type of questionnaire used; structured, semi-structured or unstructured. If structured, what type; 'rank order, matching, 'checklist', 'Likert scale'? Was it administered by self, interview or mailed? (All these are called the survey instrument/tools).

- Data analysis - Written on the method of analysis the researcher used; 'manual', or 'computer'? if statistically, is it descriptive or inferential statistics that was employed? (eg.

1% or 5% significant level?) if applicable. The type of statistical formula should also be stated.

- Data Presentation - Here the researcher discusses the various methods of data presentation, then shows the ones used; 'histogram', 'pie-chart', etc.

5. Chapter Four

The result

The various findings are presented in words or the use of tables. Various types of diagrams, plates, graphs etc should be summarized with one or two sentences that are written underneath. The result should only contain important information that is found in the table. It should only have one table at a page and no more, then give the tables titles that are adequate, clear and concise. Write the titles at the top of the page containing the table. Diagrams and plates should have their titles below them. Numbering should be in sequence and in arabic figure.

6. Chapter Five

Discussion

Here, the important findings of the study are discussed. They are compared with results of similar studies that have been mentioned in the literature review and explain the differences if there is any and why the difference(s) occurred.

7. Chapter Six

Conclusions and Recommendations

Limitations (some institutions and organizations list theirs here). State whether your study achieved its aims and

objectives or not. Then offer some recommendations to appropriate quarters based on the problems identified. The recommendations should be specific and relevant. The referencing may now come after this. The reference style should be as recommended by institution or organization and must be updated edition.

CONCLUSION

The book is arranged in three parts with seven chapters. Students at various levels of academic pursuit will benefit from the various chapters of this book. In any research project, the introduction comes after the abstract. The abstract is part of the preliminary pages. The abstract is a brief summary of the whole study. It may be written in paragraphs or in block depending on the organization preference. Under introduction, the subject matter and the methodology and findings are recorded. The statement of the problem, objective of study, assumption/hypotheses, research questions, significance of the study, the scope of the study and operational definition of terms; all come under introduction.

The introduction of the book (project/Thesis) is followed by the literature review where studies done by other researchers, which are related to the present study, are presented logically and in sequence using sub headings or paragraphs. Every author is cited in the text (citation) and referenced at the end of the study. The methodology comes in as chapter three where the research method, study area, study population, sample and sample technique, instrument for data collection, procedure for data collection, method of data analysis are documented.

Chapter four deals with the result of the findings of the study and its representation, which is based on the assumptions or research questions using percentages, tables, graphs, etc. It is then followed by the chapter on discussion. Discussion states briefly the major findings and states similarities and dissimilarities. If results are similar or not similar, reasons for such should be provided. The study is then concluded and recommendations for further studies made. It is therefore assumed that all students at various levels of academic pursuit will find this book valuable.

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GLOSSARY

- **Attribute** – A quality regarded as a natural or typical part of a subject or something.
- **Control Group** – Refers to the group whose performance on a dependent variables is used as basis for evaluation of the experimental group. A group of subjects who will not receive the treatment in the experimental study.
- **Dependent Variable** – Variable acted upon the independent variable.
- **Empirical** – past records.
- **Experimental Group** – The group/subjects that receive the treatment or manipulation in a study.
- **Hypothesis** – The statement of the relationship between two variables. The statement is predicted to be achieved.
- **Independent Variable** – That, which is the cause of the variable. It acts on the dependent variable.
- **Method** – The methodological appropriation to be adopted by the study.
- **Percentile** – A term referring to the dividing of a distribution into units from 1 to 100, to obtain their rankings.
- **Pilot Study** – A small study conducted before the major study.
- **Population** – Total number of units from which the sample is selected. Population, simply means any specified group of persons, things or measurement values.
- **Randomization** – Method by which subjects are divided or allocated into any group by random method. This method prevents bias since everybody has an equal chance of belonging to one group or the other.

- **Random Sample** - Selecting of a sample which allows every unit an equal chance of being selected.
- **Reliability** - The degree to which a tool measures a variable in the same way over a period of time.
- **Sample** - That portion of the population selected to take part in the study. It is the aggregate of a unit i.e., total number of sample units selected for a study.
- **Sampling** - Estimating population value. It is a process of selecting a fraction of the target population for inclusion in a study, to represent the entire population.
- **Sampling Fraction** - The proportion, which the sample constitutes of the total population.
- **Sampling Unit** - the individual member of a target population is a sampling unit.
- **Standard Deviation** - A statistical calculation, which shows how each score, differs from the mean.
- **Statistics** - An aggregate of units from which observations are made. It involves compilation, organization and computation of numerical data.
- **Survey** - To explore.
- **Target Population** - The population to which the research findings are applied. It may be human beings, animals or inanimate things.
- **Variable** - Any characteristic that changes from time to time.
- **Variability** - The extent to which a tool measures that which it is designed to measure.

APPENDIX

SAMPLE OF A QUESTIONNAIRE

Dear Respondent,

I am a student in the I am carrying out study on the among in Local Government Area of state.

I therefore solicit for your co-operation in answering these questions. Please note that your name is NOT required. All information provided will be treated as confidential.

Thank you
James Okafor

Name of your establishment:

Date of Birth: (If applicable)

* STRUCTURED QUESTIONNAIRE

A. Example of Demographic Data

Choose the Most Appropriate Answer

- Age**
 - 10 - 14 ()
 - 15 - 19 ()
 - 20 - 24 ()

iv. 25 & above ()

2. Educational Level/Form

i. SSS 1 ()

ii. SSS 2 ()

iii. SSS 3 ()

3. Religion

i. Anglican ()

ii. Catholic ()

iii. Moslem ()

iv. Others Specify ()

4. Marital Status

i. Married ()

ii. Not Married ()

iii. Divorced/separated ()

iv. Engaged ()

5. Occupation of Parents:

i. Civil Servant ()

ii. Businessman ()

iii. Skilled (Artisan workman) ()

iv. Others Specify ()

Structured Questionnaire for Data Collection Continued.

a. Your information on contraceptive and pregnancy is from:

(i) School mates ()

(ii) Parents ()

(iii) Teachers ()

(iv) Health Personnel ()

(v) Patent Medicine Dealers ()

(vi) Media ()

b. Adolescent Teenage Pregnancy is accompanied by:

(i) Health hazards/problems only ()

(ii) Economic Hazards/problem only ()

(iii) Social Hazards/problem only ()

(iv) All of the Above ()

C. Example of checklist format/matching.

Match the following diseases with the organisms that cause them.

Column A

Neisseria Gonococcus

Column B

A.I.D.S.

Human Immune Deficiency Virus	Thrush
Varicella virus	Syphilis
Candida albicans	Gonorrhoea
Treponema palidum	Chicken pox

D. Example of Likert format/scale

Pre-marital sex is more common among adolescent girls than boys

- Agreed
- Strongly agreed
- Not decided
- Disagree
- Strongly

E. Example of rank order scale

Arrange in order of occurrence:

A.I.D.S., Syphilis, Gonorrhea, Candidiasis Sexually

Transmitted diseases common in adolescent girls

- Candidiasis
- Gonorrhea
- A.I.D.S.
- Syphilis

UNSTRUCTURED QUESTIONNAIRE

Example:

- Sources of information on contraceptives areand
- The consequences of pre-marital sexual relationship areand
- Diarrhea in infants is mostly due to

APPENDIX II

FORMAT FOR WRITING RESEARCH PROJECTS

Before chapter one, the preliminary pages will come up which include:

- (i) The cover
- (ii) Declaration
- (iii) Dedication]
- (iv) Attestation
- (v) Acknowledgement
- (vi) Table of content
- (vii) List of tables and figures
- (viii) List of plates (optional)
- (ix) Abstract

- Chapter 1**
- Introduction
 - Introduction of Topic
 - Statement of problem/purpose of study
 - Statement of scope and limitation
 - Research question and hypothesis
 - Operational define.

- Chapter 2**
- Literature review.

- Chapter 3**
- Methodology
 - Population
 - Sample, sampling techniques
 - Research design.
 - Data collection and Analysis.
 - Limitations.

- Chapter 4**
- Data Presentation (Results)
- Chapter 5**
- Discussion of findings
 - Highlight of findings
 - Relationship with other studies/literature
- Chapter 6**
- Summary
 - Conclusion
 - Recommendation/suggestions for further studies.
 - Referencing as approved by the authority.